Lück Approximation Fails in Positive Characteristic

Hans Brende Claude Opus 5.5 (Anthropic)

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Abstract

Jaikin-Zapirain's arbitrary-field form of the Lück approximation conjecture concerns a group G, a field K, a matrix A over the group algebra $K[G]$, and the normalized ranks of A over the finite quotients G/G_i along a chain of normal subgroups of finite index with trivial intersection. It asserts that these ranks converge (part (1)) to a limit that does not depend on the chain (part (2)). We show that part (2) fails in characteristic 3 for the free group F of rank 89. There is one 110×110 matrix A over $\mathbb{F}_3[F]$ such that, over every field of characteristic 3, its normalized ranks tend to the full rank 110 along every chain with 3-group quotients that is cofinal in the pro-3 topology, while along the congruence chain of an arithmetic lattice, whose quotients are 11-groups, its normalized kernel stays at least 20, so that a subchain converges to a limit at most 90. The matrix is the Tanner operator of the Goren–Oort configuration of the Hilbert modular surface of $\mathbb{Q}(\sqrt{5})$ in characteristic 3.

The deduction is formalized in Lean 4 with Mathlib, on the standard axioms only: the criterion that separates two chains, a kernel-checked certificate, the limit 110 along pro-3-cofinal chains, convergence along chains with p-group quotients in characteristic p, and the failure of part (2) from one hypothesis. That hypothesis, that along some chain the normalized kernel of A does not tend to 0, is proved here but not formalized, by Riemann–Roch on a compactified Hilbert modular surface in characteristic 3 and the uniformization of its supersingular locus. The Lean matrix is defined from the construction itself; generated data enter only as kernel-checked certificates and caches, and as two free choices, a spanning tree and a voltage. We explain why any refutation of part (2) must prove something about infinitely many levels, and why, for this matrix, no argument that applies to all covers of the base graph can supply it; heuristically, it must use the arithmetic of congruence covers.

Contributions. The problem arose in earlier joint work of the authors on transposition on tapes [3]; Hans Brende chose to pursue it, directed the research and fixed the verification standard. Claude Opus 5.5, a large language model developed by Anthropic, found the counterexample, developed the mathematics, wrote the Lean formalization and wrote this paper under Hans Brende's direction.

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1 Introduction

The problem. Let G be a group, K a field and A an $n \times m$ matrix over the group algebra $K[G]$. For a normal subgroup N of finite index, right multiplication by the image of A is a K-linear map $\varphi_{G/N}(A)\colon K[G/N]^n \to K[G/N]^m$, and

$$\mathrm{rk}_{G/N}(A) = \frac{\dim_K \operatorname{im} \varphi_{G/N}(A)}{|G : N|}$$

is its normalized rank. A *chain* is a strictly decreasing sequence $G_0 > G_1 > \cdots$ of normal subgroups of finite index with trivial intersection. Over subfields of $\mathbb{C}$ the sequence $\mathrm{rk}_{G/G_i}(A)$ converges along every chain of a residually finite group, to a limit that does not depend on the chain: this is Lück's approximation theorem [19] and its extension to all subfields of $\mathbb{C}$ (see [12, Section 1 and Theorem 10.1]). In his survey of L^2-Betti numbers and their analogues in positive characteristic, Jaikin-Zapirain conjectures the same over an arbitrary field [12, Conjecture 1.2]: for every group G, every field K and every matrix A over $K[G]$,

(1) along every chain the sequence $\mathrm{rk}_{G/G_i}(A)$ converges, and

(2) the limit does not depend on the chain.

The survey calls parts (1) and (2) together the Lück approximation conjecture; its part (3), the strong Atiyah conjecture, is not considered here. The same two statements are parts (1) and (2) of Conjecture 2 of Jaikin-Zapirain and Souza [14]. In positive characteristic both parts are known for amenable groups [12, Section 1], [14, Section 8], and along the chains obtained from suitable embeddings into pro-p groups the limit exists and is a rank over a division ring [14, Theorem 8.1]. In a blog post Jaikin-Zapirain expects that in positive characteristic the conjecture holds only under conditions on the completion, and states a refined conjecture for embeddings into virtually pro-p groups [13].

The results. Let F be the free group of rank 89. In Section 3 we define one matrix $A \in M_{110}(\mathbb{F}_3[F])$, the Tanner operator of the Goren–Oort configuration of a Hilbert modular surface of $\mathbb{Q}(\sqrt{5})$ in characteristic 3, written over $\mathbb{F}_3$. For a field K of characteristic 3 let A_K be its image under $\mathbb{F}_3 \to K$.

- **Theorem A** (Theorem 5.6). For every field K of characteristic 3, part (2) of the conjecture fails for F over K. Hence parts (1) and (2) do not both hold for F over $\mathbb{F}_3$, and the arbitrary-field conjecture is false (Corollary 5.7).
- **Theorem P** (Theorem 4.7). Along every chain of F with 3-group quotients that is cofinal in the pro-3 topology, $\mathrm{rk}(A_K) \to 110$, the full rank.
- **Theorem H** (Theorem 6.10). Along the congruence chain $(N_k)_{k\geq 0}$ defined by a prime λ_0 above 11, whose quotients F/N_k are 11-groups of order 11^{3k},

$$\dim_{\mathbb{F}_3} \ker \varphi_{F/N_k}(A) \;\geq\; 22\,|F : N_k| - 2 \;\geq\; 20\,|F : N_k| \qquad (k \geq 0).$$

Theorem A follows from Theorems P and H by a general criterion (Proposition 5.2): for a square matrix, a full-rank limit along one chain and a normalized kernel that does not tend to 0 along another chain refute part (2). By Bolzano–Weierstrass a subchain of (N_k) then has a limit at most

90, while every pro-3-cofinal chain with 3-group quotients has limit 110. The deduction uses only that the normalized kernel of A does not tend to 0 along some chain, and for this matrix nothing weaker suffices: the limits of A_K along all chains with convergent rank sequences agree exactly when its normalized kernel tends to 0 along every chain (Section 5.3).

One further formal result puts Theorem A in context: in characteristic p the ranks of every matrix over every group converge along every chain with p-group quotients (Theorem 4.5, a known fact that we formalize), so a counterexample to part (1) must use other chains. Moreover, by Theorem P the bad chain cannot be pro-3-cofinal with 3-group quotients; the congruence chain is pro-11. This agrees with Jaikin-Zapirain's expectation, and leaves his refined conjecture for virtually pro-p completions untouched.

How to read the paper and its formal proofs. Every numbered result states its status. A line Lean: names the Lean declarations of a formalized result and their file, and the result is followed by an outline of the formal proof; the Lean statement is authoritative. A line Lean (partial): names what is formalized and what is not. Results marked Not formalized have complete proofs in the text. Remarks labelled as numerical, as proof sketches or as heuristics are not used in any proof.

Everything is formalized except Theorem H and the results used only in its proof: the lemmas of Section 6 and Lemmas 3.1, 3.3 and 3.6 of Section 3. The Lean matrix is defined by the construction of Definition 3.4 itself, from the icosians. Generated data enter only as a coset certificate and packed caches, both checked by the kernel (Lemma 3.5, `Luck.Con.blockNat_eq_cache`), and as free choices of a spanning tree, checked to be one, and of a voltage. The Lean form of Theorem A (`Luck.Op.chainIndependence_fails`) takes as an explicit hypothesis (`Luck.Op.KernelNotNegligible`) that the normalized kernel of A does not tend to 0 along some chain, which Theorem H implies; it uses no axiom beyond `propext`, `Classical.choice` and `Quot.sound`. The proof of Theorem H uses Riemann–Roch on a toroidal compactification of a Hilbert modular surface in characteristic 3, the partial Hasse invariants of Goren and the uniformization of the supersingular locus by Rapoport–Zink and Stamm; none of this is available in Mathlib. The formal statement of the conjecture (Section 2) comes with two independent checks, described there.

Related work. Lück proved the approximation theorem for L^2-Betti numbers and rational matrices [19]; Jaikin-Zapirain's survey [12] and the paper of Jaikin-Zapirain and Souza [14] develop the positive-characteristic theory through Sylvester rank functions and division rings. Monotonicity of normalized ranks along p-group extensions in characteristic p, the key to Theorem P, goes back to an argument of Bergeron, Linnell, Lück and Sauer [2, Lemma 4.1]. The geometry behind Theorem H is the Goren–Oort stratification of Hilbert modular varieties [10, 9], the description of the non-ordinary locus of Hilbert–Blumenthal surfaces by Bachmat and Goren [1], and its uniformization by Stamm and by Kudla–Rapoport [24, 17], through the Rapoport–Zink spaces [22].

Two earlier papers, *Covers and Local Generation in Linear Storage Codes* by Hans Brende and GPT-6 Astra [5] and *Bounded Nesting and Depth Reduction for Matrix Transposition on Tapes* by the present authors [3], use Jaikin-Zapirain and Souza's approximation theorem along towers of covers that come from pro-p embeddings. The second notes that along other towers, positive asymptotic rate for lifted binary codes would contradict the characteristic-2 case of the conjecture [3, Section 8.1]. Theorem A shows that in characteristic 3 such limits can depend on the tower (Section 8).

Organization. Section 2 states the conjecture and its formal version. Section 3 defines the matrix. Section 4 proves Theorem P and convergence along p-chains. Section 5 proves the criterion and

Theorem A from the hypothesis. Section 6 proves Theorem H. Section 7 explains what a proof of the hypothesis needs and why it cannot be replaced by a computation, and Section 8 lists open problems. The appendices list the Lean declarations and explain how to build and check them.

2 The conjecture and its formal statement

Throughout, G is a group, K a field, and $K[G]$ the group algebra. Vectors are row vectors and matrices act on the right, as in the survey [12, Section 1].

Definition 2.1 (Normalized rank)**.** Lean: `Luck.quotientMatrix`, `Luck.phi`, `Luck.rk` [`LuckApproximation/Statement.lean`]

Let A be an $n \times m$ matrix over $K[G]$ and N a normal subgroup of finite index. Write $\bar{A}$ for the image of A under the ring map $K[G] \to K[G/N]$ induced by $G \to G/N$. The K-linear map $\varphi_{G/N}(A)\colon K[G/N]^n \to K[G/N]^m$ is $x \mapsto x\bar{A}$, and

$$\mathrm{rk}_{G/N}(A) = \frac{\dim_K \operatorname{im} \varphi_{G/N}(A)}{|G : N|}.$$

Lemma 2.2 (Basic properties)**.** Lean: `Luck.finrank_domain`, `Luck.rk_le`, `Luck.rk_eq_sub`, `Luck.rk_one` [`LuckApproximation/Sanity.lean`]

Let N be a normal subgroup of finite index and A an $n \times m$ matrix over $K[G]$. Then $\dim_K K[G/N]^n = n|G : N|$, $\mathrm{rk}_{G/N}(A) \le n$,

$$\mathrm{rk}_{G/N}(A) = n - \frac{\dim_K \ker \varphi_{G/N}(A)}{|G : N|},$$

and the identity matrix has $\mathrm{rk}_{G/N}(I_n) = n$.

Outline of the formal proof. The group elements of G/N form a basis of $K[G/N]$. The formula for $\mathrm{rk}_{G/N}(A)$ in terms of the kernel is rank–nullity. □

We call $\dim_K \ker \varphi_{G/N}(A)/|G : N|$ the *normalized kernel* of A at N; we use it for square matrices.

Definition 2.3 (Chains)**.** Lean: `Luck.Chain`, `Luck.Chain.rkSeq` [`LuckApproximation/Statement.lean`]

A *chain* of G is a sequence $(G_i)_{i \ge 0}$ of normal subgroups of finite index that is strictly decreasing, $G_0 > G_1 > \cdots$, and has trivial intersection. Along a chain, A has the rank sequence $i \mapsto \mathrm{rk}_{G/G_i}(A)$.

A chain whose quotients G/G_i are all p-groups is a *p-chain*, for instance a 3-chain. The first term of a chain may be G itself; adding or dropping finitely many initial terms changes no limit. A group with a chain is infinite and residually finite. Requiring strict inclusions only weakens the conjecture below, so a disproof of the strict version also disproves the non-strict one (`Luck.Literal.nonStrict_imp_luck`).

Definition 2.4 (Lück approximation over a field)**.** Lean: `Luck.Convergence`, `Luck.ChainIndependence`, `Luck.LuckApproximation`, `Luck.LuckApproximationConjecture` [`LuckApproximation/Statement.lean`]

Let K be a field and G a group.

(1) *Convergence:* for every matrix A over $K[G]$ and every chain of G, the rank sequence of A converges.

(2) *Chain independence:* for every matrix A over $K[G]$ and any two chains along which the rank sequences of A converge, the two limits are equal.

Lück approximation over K for G is the conjunction of (1) and (2). The *arbitrary-field Lück approximation conjecture* [12, Conjecture 1.2(1),(2)] asserts it for every field and every group.

Part (2) as stated compares only chains along which the ranks converge, so it does not presuppose part (1). Together, (1) and (2) say that for each matrix one number is the limit along every chain (`Luck.luckApproximation_iff`). The formal target of this paper is the negation for one free group over $\mathbb{F}_3$,

$$\exists\, r,\ \neg \text{Lück approximation over } \mathbb{F}_3 \text{ for the free group of rank } r,$$

stated as `Luck.MainTheoremTarget`; it refutes the conjecture (`Luck.not_conjecture_of_target`).

Checks of the statement. The library contains two independent checks of these definitions. A second formalization, written independently from the survey's text, is proved equivalent to Definition 2.4 (`Luck.Literal.equiv`). A set of semantic checks confirms on examples what the definitions mean: $\mathrm{rk}(0) = 0$, $\mathrm{rk}_{G/N}(1-g) = 1 - 1/\mathrm{ord}(gN)$ for 1×1 matrices, convergence of $\mathrm{rk}(1-g)$ to 1 along the chain $2^i\mathbb{Z}$ for g a generator of $\mathbb{Z}$, and the fact that only infinite groups carry chains (`Luck.Checks.rk_zero`, `Luck.Checks.rk_one_sub_single`, `Luck.Checks.intChain_tendsto`, `Luck.Checks.infinite_of_chain`).

3 The Goren–Oort operator

This section defines the matrix A and relates its kernels to Tanner codes on covers of a finite graph. The arithmetic that makes these kernels large is in Section 6; here we need only the combinatorial structure.

3.1 The icosian group and the base graph

Let $E = \mathbb{Q}(\sqrt{5})$, with ring of integers $\mathcal{O}_E = \mathbb{Z}[\theta]$, $\theta = (1+\sqrt{5})/2$. The prime 3 is inert in E; we identify $\mathcal{O}_E/3$ with $\mathbb{F}_9 = \mathbb{F}_3[i]/(i^2+1)$ by $\theta \mapsto 2+i$. The prime 11 splits; let λ_0 be the prime above 11 with $\theta \equiv 4 \pmod{\lambda_0}$, so that $\mathcal{O}_E/\lambda_0^m \cong \mathbb{Z}/11^m$. Let B be the quaternion algebra over E ramified exactly at the two infinite places, and $I \subset B$ the icosian ring, a maximal order of class number 1 whose norm-one units form the binary icosahedral group $2I \cong \mathrm{SL}_2(\mathbb{F}_5)$ of order 120 [28, Proposition V.3.4 and p. 155]. Since B splits at 3, $B \otimes_E E_3 \cong M_2(\mathbb{Q}_9)$, where $\mathbb{Q}_9$ is the unramified quadratic extension of $\mathbb{Q}_3$ with ring of integers $\mathbb{Z}_9$, and

$$\Gamma^+ = I[1/3]^1/\{\pm 1\} \subset \mathrm{PSL}_2(\mathbb{Q}_9)$$

acts on the Bruhat–Tits tree $\mathcal{T}$ of $\mathrm{PGL}_2(\mathbb{Q}_9)$, which is 10-regular; it preserves the bipartition of the vertices into even and odd ones. Let Λ_0 be the vertex fixed by $I \otimes \mathcal{O}_{E,3} \cong M_2(\mathbb{Z}_9)$, let $\alpha = 1+i+j \in I$, of reduced norm 3, and put $\Lambda_1 = \alpha\Lambda_0$, a neighbour of Λ_0. The stabilizers of Λ_0, Λ_1 and the edge $e_0 = \{\Lambda_0, \Lambda_1\}$ are $A_5 = 2I/\{\pm 1\}$, $\alpha A_5 \alpha^{-1}$ and a group S_3, whose preimage $\Gamma_e \subset 2I$ is binary dihedral of order 12; and $\Gamma^+ = A_5 *_{S_3} \alpha A_5 \alpha^{-1}$ (Lemma 6.2).

For $m \geq 1$ let $\Gamma^+(\lambda_0^m) = \ker(\Gamma^+ \to \mathrm{PSL}_2(\mathcal{O}_E/\lambda_0^m))$. It is torsion-free and acts freely on $\mathcal{T}$ (Lemma 6.3). The *base graph* is $X_0 = \Gamma^+(\lambda_0)\backslash\mathcal{T}$.

Lemma 3.1 (The base graph)**.** Not formalized; a complete proof is given below.

Let $P = \mathrm{PSL}_2(\mathbb{F}_{11})$, and let H_0, H_1 and H_e be the images of A_5, $\alpha A_5 \alpha^{-1}$ and S_3 under reduction modulo λ_0. Then $H_e = H_0 \cap H_1$, and X_0 is the coset graph with even vertices P/H_0, odd vertices P/H_1 and edges P/H_e, the edge gH_e joining gH_0 to gH_1. It has 11 vertices of each parity and 110 edges, every even vertex is joined to 10 distinct odd vertices, and X_0 is the complete bipartite graph $K_{11,11}$ with a perfect matching removed. Its fundamental group is free of rank $110 - 22 + 1 = 89$.

Proof (not formalized). By Lemma 6.3, $\Gamma^+ \to P$ is onto with kernel $\Gamma^+(\lambda_0)$, which meets no vertex stabilizer. Since Γ^+ acts transitively on the even vertices, on the odd vertices and on the edges (Lemma 6.2), the even vertices of X_0 are $\Gamma^+(\lambda_0)\backslash\Gamma^+/A_5 = P/H_0$, and likewise for odd vertices and edges, with incidences as stated. This gives $660/60 = 11$ vertices of each parity and $660/6 = 110$ edges. The edges hH_e ($h \in H_0$) at the even vertex H_0 have odd ends hH_1, and two of them have the same odd end exactly when they differ by an element of $H_0 \cap H_1$. This group contains $H_e \cong S_3$, which is maximal in $H_0 \cong A_5$. If it were all of H_0, then $H_1 = H_0$, every edge at gH_0 would end at the odd vertex gH_1, and the quotient X_0 of the connected tree $\mathcal{T}$ would be disconnected. So $H_0 \cap H_1 = H_e$, and the ten edges at an even vertex end at ten distinct odd vertices. By symmetry every odd vertex is joined to ten distinct even vertices, so the missing pairs form a perfect matching. A connected graph with 22 vertices and 110 edges has free fundamental group of rank 89. □

Lifting a spanning tree of X_0 to $\mathcal{T}$ identifies $\pi_1(X_0)$ with $\Gamma^+(\lambda_0)$; we write ψ for this isomorphism (Lemma 6.4).

3.2 Local codes and data

The 10 edges at a vertex Λ correspond to the lattices Λ' with $3\Lambda \subset \Lambda' \subset \Lambda$ of index 9, that is, to the points of the projective line $\mathbb{P}^1(\Lambda/3\Lambda)$ over $\mathbb{F}_9$.

Definition 3.2 (Local codes)**.** For vectors $x_1, \dots, x_{10} \in \mathbb{F}_9^2$ representing the 10 points of $\mathbb{P}^1(\mathbb{F}_9)$, the *local code* is $C = \{(f(x_1), \dots, f(x_{10})) : f \text{ a binary quartic form over } \mathbb{F}_9\} \subseteq \mathbb{F}_9^{10}$.

Lemma 3.3 (Self-duality)**.** Not formalized; a complete proof is given below.

The local code has dimension 5 and equals its dual for the standard bilinear form, for every choice of the representatives x_j.

Proof (not formalized). A nonzero binary quartic vanishes at no more than four points of $\mathbb{P}^1$, so evaluation is injective and $\dim C = 5$. For quartics f, g put $h = fg$, of degree 8. Since $c^8 = 1$ for $c \in \mathbb{F}_9^\times$, $h(cx) = h(x)$, so $\sum_{v \in \mathbb{F}_9^2 \setminus 0} h(v) = 8 \sum_j h(x_j) = -\sum_j h(x_j)$. For a monomial $h = X^a Y^{8-a}$, $\sum_{(s,t) \in \mathbb{F}_9^2} s^a t^{8-a} = (\sum_s s^a)(\sum_t t^{8-a})$, and $\sum_{s \in \mathbb{F}_9} s^a = 0$ unless a is a positive multiple of 8 (with $s^0 = 1$, the sum for $a = 0$ is $9 = 0$). One of the two factors therefore vanishes for every $0 \le a \le 8$, and $\sum_j f(x_j) g(x_j) = 0$. So $C \subseteq C^\perp$, and equality holds because $\dim C = 5 = 10 - \dim C$. □

A *datum* on $\mathcal{T}$, or on a quotient of $\mathcal{T}$, assigns to every vertex a local code on the edges at that vertex, given by labels in the sense of Definition 3.2. Its *Tanner code* on a quotient graph X consists of the functions on the edges of X whose restriction to the edges at each vertex lies in the local code there.

Choose $p_0 \in \mathbb{F}_9^2$ spanning the image and k_0 spanning the kernel of $\alpha \bmod 3 \in M_2(\mathbb{F}_9)$, in the coordinates of $\Lambda_0/3\Lambda_0 = \mathbb{F}_9^2$. The edges at Λ_0 are $u\, e_0$ for $u \in 2I$; label $u\, e_0$ by the vector $u\, p_0$. At Λ_1, in the coordinates transported by α, label the edge $t\, e_0$, where $t = \alpha t' \alpha^{-1}$ with $t' \in 2I$, by $\mathrm{Frob}(t' k_0)$, where Frob cubes both coordinates. Two elements give the same edge exactly when they differ by an element of Γ_e, and then their labels differ by a scalar c with $c^4 = 1$: Γ_e acts on

the line of the edge through a character, which factors through the abelianization C_4 of the binary dihedral group of order 12. The space of binary quartic forms is $\mathrm{GL}_2(\mathbb{F}_9)$-stable, and $f(cx) = f(x)$ when $c^4 = 1$; so the local codes at Λ_0 and Λ_1 are well defined, and transporting them by Γ^+ gives a Γ^+-equivariant datum on $\mathcal{T}$, the *twisted datum* D_{tw}. The *untwisted datum* D_{untw} is the same without Frob. For $\Gamma \subseteq \Gamma^+(\lambda_0)$ these data descend to $\Gamma\backslash\mathcal{T}$; write $T_{\mathrm{tw}}(\Gamma\backslash\mathcal{T})$ for the Tanner code of the descent of D_{tw}. Section 6 shows that the datum that geometry produces has the type of D_{tw} or of its image under the Frobenius (Lemma 6.9), which have Tanner codes of the same dimensions.

3.3 The matrix

Fix a spanning tree τ of X_0. Its 21 edges leave 89 edges, and $F = \pi_1(X_0, \tau)$ is free on generators $x_0, \dots, x_{88}$, one for each edge outside τ, in the order of the data. Put γ_e equal to the generator of e for an edge outside τ and $\gamma_e = 1$ for an edge of τ. For an edge e, let $m_0(e) = (x^{4-a}y^a)_{a=0,\dots,4}$ be the quartic monomials evaluated at the label of e at its even end, and $m_1(e)$ the same at its odd end, with the twisted labels.

Definition 3.4 (The operator)**.** Lean: `Luck.Op.F`, `Luck.Op.gamma`, `Luck.Op.entryEdge`, `Luck.Op.coupling`, `Luck.Op.opA` [`LuckApproximation/Operator.lean`]

The 55×55 matrix A_9 over $\mathbb{F}_9[F]$ has rows indexed by pairs (v_0, c) of an even vertex and $c \in \{0, \dots, 4\}$ and columns by pairs (v_1, r) of an odd vertex and $r \in \{0, \dots, 4\}$, and

$$A_9[(v_0, c), (v_1, r)] = \sum_{e\colon v_0 — v_1} m_1(e)_r \, m_0(e)_c \, \gamma_e \in \mathbb{F}_9[F].$$

By Lemma 3.1 each entry has at most one term. The matrix $A \in M_{110}(\mathbb{F}_3[F])$ is the restriction of scalars of A_9 along $\mathbb{F}_9 = \mathbb{F}_3[i]/(i^2+1)$: an entry $c_0 + c_1 i$ becomes the block $\left(\begin{smallmatrix} c_0 & c_1 \\ -c_1 & c_0 \end{smallmatrix}\right)$, row (v_0, c), component a, has index $10v_0 + 2c + a$, and column (v_1, r), component b, has index $10v_1 + 2r + b$. For a field K of characteristic 3, A_K is the image of A under $\mathbb{F}_3 \to K$ (`Luck.Op.opK`).

In Lean the matrix is this construction, computed on natural numbers (`LuckApproximation/Construction.lean`). The 120 icosians are given by the standard formula (`Luck.Con.icosians`): the units $\pm1, \pm i, \pm j, \pm k$, the elements $(\pm1 \pm i \pm j \pm k)/2$, and the elements whose coordinates are an even permutation of $(0, \pm\frac{1}{2}, \pm\frac{\theta^{-1}}{2}, \pm\frac{\theta}{2})$. They are reduced by the splittings $i \mapsto \left(\begin{smallmatrix} 1 & 1 \\ 1 & -1 \end{smallmatrix}\right)$, $j \mapsto \left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$ over $\mathbb{F}_9$ and $i \mapsto \left(\begin{smallmatrix} 1 & 3 \\ 3 & -1 \end{smallmatrix}\right)$, $j \mapsto \left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$ over $\mathbb{F}_{11}$ (`Luck.Con.red3`, `Luck.Con.red11`), and the labels, the quartic monomials and the blocks are computed from them (`Luck.Con.label0`, `Luck.Con.label1`, `Luck.Con.mono`, `Luck.Con.blockNat`). Generated data enter only as a coset certificate and packed caches, both checked by the kernel, and as free choices. The certificate (`ConData.lean`) gives coset representatives of the vertices, the position icosians, and the positions of each edge at its two ends; Lemma 3.5 checks it. Packed copies of the incidences and blocks speed up the certificate of Section 4.1; the kernel checks that they agree with the construction (`Luck.Con.blockNat_eq_cache`, `Luck.Con.edgeOfNat_eq_cache`), so nothing in the proof trusts them. The choices are the spanning tree, with a numbering of the other edges, which Lemma 3.5 checks to be a spanning tree, and the voltage χ of Section 4.1. The coset representatives are also a choice: they fix coordinates at each vertex. Another choice multiplies the labels at a vertex by a fixed matrix in the image of $2I$, and so multiplies its five rows (or columns) of A_9 by a fixed invertible 5×5 matrix over $\mathbb{F}_9$, which changes no kernel dimension; another choice of position icosians multiplies labels by scalars c with $c^4 = 1$ and leaves the matrix unchanged.

Lemma 3.5 (The Lean enumeration)**.** Lean: `Luck.Con.splittings_ok`, `Luck.Con.subgroups_ok`, `Luck.Con.cosetGraph_ok`, `Luck.Con.spanningTree_ok`, `Luck.Con.labels_ok` [`LuckApproximation/ConChecks.lean`]

Let $\pm H_0$ and $\pm H_1$ be the sets of images in $\mathrm{SL}_2(\mathbb{F}_{11})$ of the 120 icosians and of their conjugates by α, and of their negatives, and $\pm H_e = \pm H_0 \cap \pm H_1$. Then:

(a) the two splittings respect $i^2 = j^2 = -1$ and $ij = -ji$, and α is invertible modulo λ_0;

(b) $\pm H_0$ and $\pm H_1$ consist of matrices of determinant 1, and the product of any two of the 120 images lies in $\pm H_0$ (resp. $\pm H_1$); so $\pm H_0$ and $\pm H_1$ are subgroups of $\mathrm{SL}_2(\mathbb{F}_{11})$;

(c) the representatives of the even vertices have determinant 1 and lie in distinct cosets of $\pm H_0$ that cover $\mathrm{SL}_2(\mathbb{F}_{11})$, and likewise the representatives of the odd vertices and $\pm H_1$;

(d) the ten position icosians at the even vertices, and the conjugates by α of the ten position icosians at the odd vertices, represent the ten cosets of $\pm H_e$ in $\pm H_0$ (resp. $\pm H_1$) once each;

(e) the ends and positions of the edges are in range, no two edges have the same ends, the pairs of an end and a position are distinct, and each edge is the same coset of $\pm H_e$ seen from its two ends;

(f) the edges without a generator form a spanning tree, and the other edges carry distinct generators;

(g) $p_0 \neq 0$, $k_0 \neq 0$ and $(\alpha \bmod 3)\, k_0 = 0$; since p_0 is a column of $\alpha \bmod 3$, it spans the image and k_0 the kernel.

Outline of the formal proof. Each statement is a Boolean computation on natural numbers, evaluated by the kernel (`decide +kernel`). Matrices over $\mathbb{F}_{11}$ are coded as single numbers and the sets $\pm H_0$, $\pm H_1$ as bit masks, which are closed under negation by construction; coverage is checked over all 11^4 matrices. □

By (b), $\pm H_0$ and $\pm H_1$ are subgroups, the preimages in $\mathrm{SL}_2(\mathbb{F}_{11})$ of H_0 and H_1, and by Lemma 3.1 $\pm H_e$ is the preimage of H_e. So (c)–(e) say that the enumeration is the coset graph of Lemma 3.1, and (g) that the labels are those of Section 3.2. So the Lean operator `Luck.Op.opA` is the matrix of Definition 3.4, with no step resting on generated data.

Lemma 3.6 (Kernels are Tanner codes of covers)**.** Not formalized; a complete proof is given below.

Let N be a normal subgroup of F of finite index, and let $X_N \to X_0$ be the regular cover with group F/N and voltages $\gamma_e N$, with the datum lifted from X_0. Then

$$\dim_{\mathbb{F}_3} \ker \varphi_{F/N}(A) = 2 \dim_{\mathbb{F}_9} T_{\mathrm{tw}}(X_N).$$

Proof (not formalized). Restriction of scalars commutes with passing to F/N, and it doubles dimensions of kernels, so it suffices to show $\dim_{\mathbb{F}_9} \ker \varphi_{F/N}(A_9) = \dim_{\mathbb{F}_9} T_{\mathrm{tw}}(X_N)$. The vertices of X_N are the pairs (v, g) with $g \in F/N$, and an edge e of X_0 from v_0 to v_1 lifts to the edges from (v_0, g) to $(v_1, g\gamma_e)$. A row vector $x \in \mathbb{F}_9[F/N]^{55}$ assigns to each even vertex (v_0, g) of X_N the quartic form $f_{v_0,g}$ whose coefficients are the coefficients $x_{(v_0,c)}(g)$ of g in $x_{(v_0,c)}$, $c = 0, \ldots, 4$. Let w be the function on the edges of X_N whose value on an edge is the value of the form at its even end at the label of the edge. For $h \in F/N$, the coefficient of h in $(x\bar{A}_9)_{(v_1,r)}$ is

$$\sum_e m_1(e)_r \sum_c m_0(e)_c \, x_{(v_0,c)}(h\gamma_e^{-1}) = \sum_e m_1(e)_r \, w(\tilde{e}),$$

where e runs over the edges of X_0 at v_1, v_0 is the even end of e, and $\tilde{e}$ is the lift of e with odd end (v_1, h), whose even end is $(v_0, h\gamma_e^{-1})$. So it is the r-th parity check at the odd vertex (v_1, h).

The map $x \mapsto w$ is injective because evaluation of quartics at ten points is injective (Lemma 3.3), and its image is the set of edge functions whose restriction to every even vertex lies in the local code. The vectors $(m_1(e)_r)_e$, $r = 0, \dots, 4$, span the local code at the odd vertex, which is self-dual (Lemma 3.3); so the parity checks at (v_1, h) vanish exactly when the restriction of w to its edges lies in the local code there. Hence $\ker \varphi_{F/N}(A_9) \cong T_{\mathrm{tw}}(X_N)$. □

4 Chains with 3-group quotients

This section proves Theorem P: along chains with 3-group quotients that are cofinal in the pro-3 topology, the normalized rank of A_K tends to the full rank. One finite certificate controls the cyclic covers F/U_j defined by one homomorphism, and monotonicity of ranks along 3-group extensions carries the bound to every chain that passes below them.

4.1 The certificate

The data fix a homomorphism $\chi\colon F \to \mathbb{Z}/3$ on the generators, with $\chi(x_0) = 1$ for the first generator. For an edge e let $v(e) \in \{0, 1, 2\}$ represent $\chi(\gamma_e)$, and let $M_e \in M_{110}(\mathbb{F}_3)$ be the block of e in A, so that $A = \sum_e M_e \gamma_e$. Put

$$B_0 = \sum_e M_e, \qquad B_1 = \sum_e v(e) M_e, \qquad B_2 = \sum_{e\colon v(e)=2} M_e.$$

Under $F \to \mathbb{Z}/3 = \langle t \rangle$, $\mathbb{F}_3[\mathbb{Z}/3] = \mathbb{F}_3[u]/(u^3)$ with $u = t - 1$, and $t^v = (1 + u)^v$ for $v \in \{0, 1, 2\}$, so the image of A is $B_0 + uB_1 + u^2 B_2$. For a field K of characteristic 3 write B_i also for the images in $M_{110}(K)$.

Lemma 4.1 (The certificate)**.** Lean: `Luck.Op.hW`, `Luck.Op.hb`, `Luck.Op.hy`, `Luck.Op.hC`, `Luck.Op.hCinv`, `Luck.Op.finrank_ker_B0_K`, `Luck.Op.connecting_K` [`LuckApproximation/Level1.lean`]

Let K be a field of characteristic 3.

(a) $\dim_K \ker(x \mapsto xB_0) \le 42$.

(b) If $x, z \in K^{110}$ satisfy $xB_0 = 0$ and $xB_1 = zB_0$, then $x = 0$.

Outline of the formal proof. The certificate consists of matrices over $\mathbb{F}_3$: a 68×68 submatrix $B_0[R, S]$ with an inverse W; 42 vectors b_a with $b_a B_0 = 0$ and 42 vectors y_k with $B_0 y_k = 0$; and the matrix $C = (b_a B_1 y_k)_{a,k}$ with an inverse. Their identities are checked by the Lean kernel (`decide +kernel`). The rows of the right-hand factors, with entries in $\{0, 1, 2\}$, are packed in base 512, so that the product of a row x with a matrix becomes one natural number $\sum_k x_k \hat{Y}_k$, whose base-512 digits are too small to carry (Kronecker substitution, `LeanMachine.PackedCheck.prodOK_sound`). Mapping the identities along $\mathbb{F}_3 \to K$, the invertible minor gives (a) (`LeanMachine.KernelCertificate.finrank_ker_add_le_of_minor`). For (b), the b_a span $\ker(x \mapsto xB_0)$, because they are independent (C is invertible) and the kernel has dimension at most 42; if $x = \sum_a \lambda_a b_a$ and $xB_1 = zB_0$, then $\sum_a \lambda_a C_{ak} = xB_1 y_k = zB_0 y_k = 0$ for every k, so $\lambda = 0$ (`LeanMachine.KernelCertificate.connecting_injective`). □

Remark 4.2 (Numerical)**.** At $N = F$ the matrix $\varphi_{F/F}(A)$ is B_0. The certificate contains 42 vectors b_a with $b_a B_0 = 0$, independent because C is invertible, so $\dim_{\mathbb{F}_3} \ker B_0 = 42$, and $\dim_{\mathbb{F}_9} T_{\mathrm{tw}}(X_0) = 21$ by Lemma 3.6. The untwisted datum gives 20 (`python3tools/icosian_graph.py11twuntw`).

The certificate is the only computation in the proof of Theorem P. Part (b) says that the connecting map $\ker B_0 \to \operatorname{coker} B_0$, $x \mapsto [xB_1]$, is injective; the next lemma turns this into a bound at infinitely many levels.

4.2 Cyclic 3-power covers

For $j \geq 0$ let $\chi_j \colon F \to \mathbb{Z}/3^j$ send the generator x_e to $v(e)$, read modulo 3^j, and let $U_j = \ker \chi_j$. Since $\chi_j(x_0) = 1$, χ_j is onto, and F/U_j is cyclic of order 3^j.

Lemma 4.3 (Cyclic covers)**.** Lean: `Luck.Op.chiPow`, `Luck.Op.Ncyc`, `Luck.Op.index_Ncyc`, `Luck.Op.isPGroup_Ncyc`, `Luck.Op.finrank_ker_Ncyc_le`, `Luck.Op.rk_Ncyc` [`LuckApproximation/CyclicTower.lean`]

Let K be a field of characteristic 3 and $j \geq 0$. Then $|F : U_j| = 3^j$, F/U_j is a 3-group, $\dim_K \ker \varphi_{F/U_j}(A_K) \leq 42$, and

$$\operatorname{rk}_{F/U_j}(A_K) \;\geq\; 110 - \frac{42}{3^j}.$$

Outline of the formal proof. Let $M = 3^j$ and let t generate $\mathbb{Z}/M$. In characteristic 3, $(t-1)^M = t^M - 1 = 0$, so with $u = t - 1$ the powers u^k, $k < M$, span $K[\mathbb{Z}/M]$ and are linearly independent (`LeanMachine.CyclicAlg.span_eq_top`, `LeanMachine.CyclicAlg.linIndep`). Since each $v(e) \in \{0, 1, 2\}$, the image of A_K in $K[\mathbb{Z}/M]$ is $B_0 + uB_1 + u^2 B_2$ with the same B_i for every j. In the basis u^k, right multiplication by it sends $x = \sum_k x_k u^k$ to the vector with coordinates $x_k B_0 + x_{k-1} B_1 + x_{k-2} B_2$. If this vanishes, then $x_0 B_0 = 0$ and $x_0 B_1 = -x_1 B_0$, so $x_0 = 0$ by Lemma 4.1(b); repeating the argument gives $x_k = 0$ for $k < M - 1$, so every kernel vector is $u^{M-1} b$ with $b B_0 = 0$ (`LeanMachine.CyclicAlg.finrank_ker_le`). Hence the kernel has dimension at most $\dim \ker B_0 \leq 42$, and Lemma 2.2 gives the rank bound. □

4.3 Monotonicity and chains

Lemma 4.4 (Monotonicity)**.** Lean: `Luck.rk_le_rk_of_isPGroup` [`LuckApproximation/Monotone.lean`]

Let K be a field of characteristic p, G a group, $N' \leq N$ normal subgroups of finite index with G/N' a p-group, and A a matrix over $K[G]$. Then $\operatorname{rk}_{G/N}(A) \leq \operatorname{rk}_{G/N'}(A)$.

Outline of the formal proof. The lemma is a lemma of Bergeron, Linnell, Lück and Sauer [2, Lemma 4.1], applied to the cokernel of A. We follow the argument of Kionke [16, Lemma 2.6 and Theorem 2.5], peeling off one central subgroup of order p at a time. By induction on $|G : N'|$, it suffices to treat $N' \leq M$ with M/N' generated by a central element ζ of order p in G/N'; such a ζ exists in the image of N because a nontrivial normal subgroup of a p-group meets its centre. Put $s = \zeta - 1 \in K[G/N']$. Then $s^p = \zeta^p - 1 = 0$, and the kernel of $K[G/N'] \to K[G/M]$ lies in $sK[G/N']$. If $R' \subseteq V' = K[G/N']^m$ is the image of $\varphi_{G/N'}(A)$, then the spaces $R' + s^i V'$, $i = 0, \dots, p$, descend from V' to R', and each step has dimension at most $\dim V'/(R' + sV') \leq m|G : M| - \dim \operatorname{im} \varphi_{G/M}(A)$. Hence $m|G : N'| - \dim R' \leq p\,(m|G : M| - \dim \operatorname{im} \varphi_{G/M}(A))$, which is the claim because $|G : N'| = p|G : M|$. □

Theorem 4.5 (Convergence along p-chains)**.** Lean: `Luck.exists_tendsto_of_isPGroup` [`LuckApproximation/Monotone.lean`]

Let K be a field of characteristic p and (G_i) a chain of a group G whose quotients G/G_i are p-groups. For every matrix A over $K[G]$ the rank sequence $\operatorname{rk}_{G/G_i}(A)$ converges.

Outline of the formal proof. By Lemma 4.4 the sequence is non-decreasing, and it is bounded by Lemma 2.2. □

This is known [12, Proposition 11.2], as a consequence of [2, Lemma 4.1]; we include it because its formal proof is short.

So part (1) of the conjecture holds along every chain with p-group quotients in characteristic p; a counterexample to part (1) would have to use other chains. The chains of Theorem P are of this kind.

Lemma 4.6 (p-chains below given subgroups)**.** Lean: `Luck.exists_pChain_le` [`LuckApproximation/PChain.lean`]

Let p be a prime, $r \geq 1$, and $(M_j)_{j\geq 0}$ normal subgroups of finite index of the free group of rank r whose quotients are p-groups. There is a chain (G_j) of the free group whose quotients are p-groups and with $G_j \leq M_j$ for all j.

Outline of the formal proof. Free groups of finite rank are residually finite p-groups (`LeanMachine.freeGroup_residually_pGroup`; the Lean proof uses unitriangular matrices over $\mathbb{Z}/p^k$, a matrix form of Magnus' embedding). Below every normal subgroup of p-power index there is a strictly smaller one (`LeanMachine.ResidualP.exists_goodP_lt`). Put $G_0 = M_0$, enumerate the group as $g_0, g_1, \ldots$, and choose G_{j+1} of p-power index strictly below $G_j \cap M_{j+1}$ and, if $g_j \neq 1$, not containing g_j; then the intersection is trivial. □

Theorem 4.7 (Theorem P: the full limit along pro-3-cofinal chains)**.** Lean: `Luck.Op.tendsto_full_of_cofinal`, `Luck.Op.ProThreeCofinal`, `Luck.Op.tendsto_full_of_proThreeCofinal`, `Luck.Op.exists_chain_tendsto_full` [`LuckApproximation/Main.lean`]

Let K be a field of characteristic 3 and (G_i) a chain of F whose quotients are 3-groups.

(a) If every U_j contains some G_i, then $\mathrm{rk}_{F/G_i}(A_K) \to 110$.

(b) In particular $\mathrm{rk}_{F/G_i}(A_K) \to 110$ if the chain is cofinal in the pro-3 topology, that is, if every normal subgroup of 3-power index contains some G_i.

(c) There is a chain as in (a): a chain of F with 3-group quotients and $G_j \leq U_j$ for all j, along which $\mathrm{rk}(A_K) \to 110$.

Outline of the formal proof. (a) The ranks are at most 110. Given $a < 110$, choose j with $42/3^j < 110 - a$ and i_0 with $G_{i_0} \leq U_j$. For $i \geq i_0$ we have $G_i \leq U_j$ and F/G_i is a 3-group, so Lemmas 4.3 and 4.4 give $\mathrm{rk}_{F/G_i}(A_K) \geq \mathrm{rk}_{F/U_j}(A_K) \geq 110 - 42/3^j > a$. (b) Each U_j is normal of 3-power index. (c) Lemma 4.6 applied to the sequence (U_j) gives such a chain (`Luck.exists_pChain_le`); `Luck.Op.exists_chain_tendsto_full` records the chain with these properties and its limit. □

Theorem P is the side of Theorem A that a computation can reach: one certificate and two uniform arguments give the limit at infinitely many levels. The other side, Theorem H, has no known proof of this kind (Section 7).

5 The criterion and the main theorem

5.1 A criterion for the failure of part (2)

Lemma 5.1 (Subchains and frequent drops)**.** Lean: `Luck.Chain.comp`, `Luck.not_chainIndependence_of_frequently` [`LuckApproximation/Chains.lean`]

A subsequence of a chain is a chain. Let A be an $n \times m$ matrix over $K[G]$ and let (G_i), (G'_i) be chains of G. If $\mathrm{rk}_{G/G_i}(A) \to l$ and $\mathrm{rk}_{G/G'_i}(A) \le l - \varepsilon$ for infinitely many i, for some $\varepsilon > 0$, then part (2) fails for G over K.

Outline of the formal proof. The infinitely many indices with $\mathrm{rk}_{G/G'_i}(A) \le l - \varepsilon$ form a subchain. Its ranks lie in $[0, n]$ (Lemma 2.2), so by Bolzano–Weierstrass a further subchain converges, to a limit at most $l - \varepsilon$. Two chains then have convergent rank sequences with different limits. □

Proposition 5.2 (The criterion)**.** Lean: `Luck.Chain.kerSeq`, `Luck.Chain.rkSeq_eq_sub`, `Luck.not_chainIndependence_of_full` [`LuckApproximation/Chains.lean`]

Let A be an $n \times n$ matrix over $K[G]$, and let (G_i), (G'_i) be chains of G. If $\mathrm{rk}_{G/G_i}(A) \to n$ and the normalized kernel $\dim_K \ker \varphi_{G/G'_i}(A)/|G : G'_i|$ does not tend to 0, then part (2) fails for G over K.

Outline of the formal proof. The normalized kernel is non-negative, so if it does not tend to 0 it is at least some $\varepsilon > 0$ infinitely often. There $\mathrm{rk}_{G/G'_i}(A) \le n - \varepsilon$ (Lemma 2.2), and Lemma 5.1 applies with $l = n$. □

The criterion is general: it needs one chain on which a square matrix becomes asymptotically injective and one on which it does not.

Inner rank. If $A = PQ$ with P of size $n \times r$, then $\varphi_{G/N}(A)$ factors through $K[G/N]^r$, so $\mathrm{rk}_{G/N}(A) \le r$ at every level. Hence every limit of a rank sequence is at most the inner rank of A, the least such r. For a free group, the inner rank over $K[F]$ equals the rank over Cohn's universal division ring of fractions [7, 14]. For the matrix of Section 3, Theorem P gives the limit 110 along some chain, so its inner rank is 110: the first chain of the criterion is the best possible, and the second is the whole difficulty.

5.2 Base change

Lemma 5.3 (Kernels grow under field extension)**.** Lean: `Luck.quotientMatrix_map`, `Luck.finrank_ker_le_map` [`LuckApproximation/BaseChange.lean`]

Let $k \to K$ be a field extension, A a matrix over $k[G]$ and A_K its image over $K[G]$. For every normal subgroup N of finite index, $\dim_k \ker \varphi_{G/N}(A) \le \dim_K \ker \varphi_{G/N}(A_K)$.

Outline of the formal proof. Mapping coefficients commutes with passing to G/N. A k-basis of the kernel stays in the kernel over K, and stays linearly independent: in the basis of group elements its coordinates are vectors over k, and a family over k is linearly independent over K if it is over k (Mathlib's `linearIndependent_algebraMap_comp_iff`). □

Equality holds, since the rank of a matrix over a field does not change under extension, but only the inequality enters the proofs: it carries a large kernel over $\mathbb{F}_3$ to every field of characteristic 3.

5.3 The hypothesis and Theorem A

Definition 5.4 (The hypothesis)**.** Lean: `Luck.Op.kerDensity`, `Luck.Op.KernelNotNegligible`, `Luck.Op.KernelDensityHypothesis` [`LuckApproximation/Hypothesis.lean`]

For a chain (G_i) of F, the normalized kernel of A at level i is $\dim_{\mathbb{F}_3} \ker \varphi_{F/G_i}(A)/|F : G_i|$ (`Luck.Op.kerDensity`). The hypothesis *(KN)* is that along some chain of F the normalized kernel of A does not tend to 0. The hypothesis *(KD)* is that along some chain there is $\varepsilon > 0$ with $\varepsilon |F : G_i| \le \dim_{\mathbb{F}_3} \ker \varphi_{F/G_i}(A)$ for every i.

Lemma 5.5. Lean:
`Luck.Op.kernelNotNegligible_of_density` [`LuckApproximation/Hypothesis.lean`]
(KD) implies (KN).

Outline of the formal proof. A sequence bounded below by $\varepsilon > 0$ does not tend to 0. □

Theorem H (Section 6) proves (KD) with $\varepsilon = 20$ along the congruence chain, and hence (KN).

Theorem 5.6 (Theorem A)**.** Lean (partial): `Luck.Op.kerDensity_le`, `Luck.Op.chainIndependence_fails`, `Luck.Op.CharThreeChainTarget`, `Luck.Op.charThreeChainTarget`, `Luck.Op.charThreeChainTarget_of_density` [`LuckApproximation/Main.lean`]
Not formalized: the hypothesis (KN), an explicit hypothesis of the Lean theorem, which follows from Theorem H (Theorem 6.10).
For every field K of characteristic 3, part (2) of the conjecture fails for the free group F of rank 89 over K.

Outline of the formal proof. By Theorem P there is a chain along which $\mathrm{rk}(A_K) \to 110$. Let (G_i') be a chain as in (KN). By Lemma 5.3, the normalized kernel of A_K over K is at least that of A over $\mathbb{F}_3$, so it does not tend to 0 either (`Luck.Op.kerDensity_le`). Proposition 5.2 applies to A_K. □

The proof gives more. With Theorem H, the rank sequence of A_K is at most $110 - 20 = 90$ along the whole congruence chain, so a subchain converges to a limit at most 90, while every pro-3-cofinal chain with 3-group quotients has limit 110 (Theorem P). The Lean theorem states the conclusion as `Luck.ChainIndependence` failing, for every K (`Luck.Op.CharThreeChainTarget`).

Corollary 5.7 (The conjecture is false)**.** Lean (partial): `Luck.Op.luckApproximation_fails`, `Luck.Op.mainTheorem`, `Luck.Op.conjecture_false` [`LuckApproximation/Main.lean`]
Not formalized: the hypothesis (KN), which follows from Theorem H.
Parts (1) and (2) do not both hold for F over $\mathbb{F}_3$. Hence the target $\exists\, r,\ \neg$ Lück approximation over $\mathbb{F}_3$ for the free group of rank r holds, and the arbitrary-field Lück approximation conjecture is false.

Outline of the formal proof. Theorem A with $K = \mathbb{F}_3$ refutes part (2), hence the conjunction; `Luck.not_conjecture_of_target` gives the last claim. □

Corollary 5.8 (Unconditional form)**.** Lean: `Luck.Op.kernel_density_tendsto_zero`, `Luck.Op.kernel_density_tendsto_zero_of_luck` [`LuckApproximation/Main.lean`]
If part (2) holds for F over some field of characteristic 3, then the normalized kernel of A over $\mathbb{F}_3$ tends to 0 along every chain of F. The same holds if parts (1) and (2) hold over $\mathbb{F}_3$.

Outline of the formal proof. Otherwise (KN) holds along some chain, and Theorem A refutes the assumption. □

Conversely, if (KN) fails, then the ranks of A tend to 110 along every chain (Lemma 2.2), and by equality in Lemma 5.3 so do those of A_K. So for this matrix the limits along all chains with convergent rank sequences agree exactly when its normalized kernel tends to 0 along every chain, and (KN) cannot be weakened. The formal and informal parts meet in this one statement about one matrix: Corollary 5.8 is proved in Lean without hypotheses, and Theorem H, proved in the next section, contradicts its conclusion along the congruence chain.

6 Proof of Theorem H

This section is not formalized. We keep the notation of Section 3: $E = \mathbb{Q}(\sqrt{5})$, $\mathcal{O}_E = \mathbb{Z}[\theta]$, the icosian ring I, the group $\Gamma^+ = I[1/3]^1/\{\pm 1\}$ acting on the tree $\mathcal{T}$ of $\mathrm{PGL}_2(\mathbb{Q}_9)$, and the prime $\lambda_0 \mid 11$. We use that θ has norm -1, that E has narrow class number one, and that every totally positive unit of $\mathcal{O}_E$ is a square.

Theorem H (Theorem 6.10) is proved in four steps. (H1) The group Γ^+ is an amalgam of two copies of A_5 (Lemmas 6.1 and 6.2), its congruence subgroups form a chain (Lemma 6.3), and the kernels of A along that chain are Tanner codes of quotients of $\mathcal{T}$ (Lemmas 3.6 and 6.4). (H2) Riemann–Roch on a Hilbert modular surface in characteristic 3 gives many sections of a line bundle on its Goren–Oort divisor (Lemma 6.5). (H3) That divisor is a quotient of a union of projective lines indexed by $\mathcal{T}$, on which the line bundle has degree 4 (Lemma 6.6), so its sections form a Tanner code (Lemma 6.7). (H4) This Tanner code is the one of the twisted datum (Lemmas 6.8 and 6.9). Steps (H2) and (H3) rest on published theorems, listed where they are used. The new arguments are Lemma 6.8, the degree computation in Lemma 6.6, Lemma 6.9, and the assembly in Section 6.5.

6.1 The group, the chain and the covering

Let σ be the Frobenius of $\mathbb{F}_9$, and also its lift to $\mathbb{Q}_9$.

Lemma 6.1 (A_5 on the projective line)**.** Not formalized; a complete proof is given below.
Let A_5 act faithfully on $\mathbb{P}^1$ over $\bar{\mathbb{F}}_3$.

(a) The action comes from an irreducible faithful representation of degree 2 of the Schur cover $2I \cong \mathrm{SL}_2(\mathbb{F}_5)$ on which -1 acts by -1. Up to isomorphism there are exactly two such representations; they are defined over $\mathbb{F}_9$, are not isomorphic, and are exchanged by σ.

(b) The centralizer of A_5 in $\mathrm{PGL}_2(\bar{\mathbb{F}}_3)$ is trivial.

(c) The orbits of A_5 on $\mathbb{P}^1(\bar{\mathbb{F}}_3)$ have sizes 10, 12 and 60 only. The orbit of size 10 consists of the fixed points of the ten subgroups S_3 of A_5; each of them fixes exactly one point, and S_3 is self-normalizing in A_5.

Proof (not formalized). (a) A faithful action of A_5 on $\mathbb{P}^1$ is a faithful projective representation of degree 2. The group A_5 has no faithful representation of degree 2 in characteristic 3 (its 3-modular irreducible representations have degrees $1, 3, 3, 4$ [15]), so the action lifts to a faithful representation of $2I$ on which -1 acts by -1. It is irreducible, since a reducible one has image in a Borel subgroup, which is solvable. The faithful 3-modular irreducible representations of $\mathrm{SL}_2(\mathbb{F}_5)$ have degrees $2, 2, 6$ [15], so there are exactly two of degree 2. Their traces on an element of order 5 are the two roots of $x^2 + x - 1$ in $\mathbb{F}_9$, so they are defined over $\mathbb{F}_9$, not isomorphic, and exchanged by σ.

(b) If $g \in \mathrm{PGL}_2(\bar{\mathbb{F}}_3)$ commutes with the image of A_5, and ρ is a representation as in (a), then $g\rho(h)g^{-1} = c(h)\rho(h)$ for a character c of $2I$, which is trivial because $2I$ is perfect; so g commutes with the absolutely irreducible $\rho(2I)$ and is a scalar.

(c) The stabilizer of a point lies in a Borel subgroup, whose finite subgroups are extensions of a cyclic group of order prime to 3 by a 3-group. Among the subgroups of A_5 only 1, C_2, C_3, C_5 and S_3 have this form. An element of order 3 is unipotent and has one fixed point, which is fixed by its normalizer S_3; so each S_3 fixes exactly that point, and two different subgroups S_3 fix different points, because together they generate A_5, which fixes no point. The ten subgroups S_3 are conjugate, so their fixed points form one orbit of size 10. Each of these points is fixed by the three

involutions of its S_3, which accounts for all $15 \cdot 2$ fixed points of the fifteen involutions; so no point has stabilizer C_2. An element of order 5 is semisimple with two fixed points, whose stabilizer is C_5, since the dihedral group of order 10 is not of the form above. This leaves orbits of sizes 10, 12 and 60. Finally S_3 is maximal and not normal in A_5, hence self-normalizing. □

Lemma 6.2 (The amalgam)**.** Not formalized; a complete proof is given below.

Γ^+ acts on $\mathcal{T}$ without inversions, transitively on the even vertices, on the odd vertices and on the edges. The stabilizers of Λ_0, Λ_1 and e_0 are $A_5 = 2I/\{\pm 1\}$, $\alpha A_5 \alpha^{-1}$ and a subgroup S_3, whose preimage $\Gamma_e \subset 2I$ is binary dihedral of order 12. Hence $\Gamma^+ = A_5 *_{S_3} \alpha A_5 \alpha^{-1}$, and Γ^+ is perfect.

Proof (not formalized). Elements of reduced norm 1 preserve the parity of vertices. A vertex of $\mathcal{T}$ is the class of a $\mathbb{Z}_9$-lattice in $\mathbb{Q}_9^2$; by the local–global correspondence for lattices [28, Section III.5.A] it is the class of the completion at 3 of a right I-ideal $\mathfrak{a}$ with $\mathfrak{a}_\mathfrak{q} = I_\mathfrak{q}$ for all primes $\mathfrak{q} \nmid 3$, and such an $\mathfrak{a}$ is principal, $\mathfrak{a} = \beta I$ with $\beta \in I[1/3]^\times$, because I has class number one ([28, p. 155], the case $m = 5$, $D = N = 1$ of the table there). The reduced norm of β is a totally positive unit of $\mathcal{O}_E[1/3]$, so it is $3^a\theta^{2b}$; replacing β by $\beta\theta^{-b}$ we may take $b = 0$. If a is even, $\beta/3^{a/2} \in \Gamma^+$ moves Λ_0 to the vertex of $\mathfrak{a}$; if a is odd, $(\beta/3^{(a-1)/2})\alpha^{-1}$ moves Λ_1 to it. So Γ^+ is transitive on the vertices of each parity. The stabilizer of Λ_0 is the image of $I^1 = 2I$. Reduction modulo 3 is injective on the finite group $2I$, because $1 + 3M_2(\mathbb{Z}_9)$ is torsion-free, so A_5 acts faithfully on the 10 points of $\mathbb{P}^1(\Lambda_0/3\Lambda_0) = \mathbb{P}^1(\mathbb{F}_9)$, which correspond to the edges at Λ_0. By Lemma 6.1(c) they form one orbit, with stabilizer S_3, whose preimage in $2I \cong \mathrm{SL}_2(\mathbb{F}_5)$ is binary dihedral of order 12. Hence Γ^+ is transitive on edges, and Bass–Serre theory [23, Section I.4] gives the amalgam. As a check, the Euler characteristic $22 - 110 = -88$ of X_0 equals $660\,(1/60 + 1/60 - 1/6)$. An amalgam of perfect groups is perfect. □

Lemma 6.3 (The chain)**.** Not formalized; a complete proof is given below.

For $m \geq 1$ the map $\Gamma^+ \to \mathrm{PSL}_2(\mathcal{O}_E/\lambda_0^m) = \mathrm{PSL}_2(\mathbb{Z}/11^m)$ is onto, and $\Gamma^+(\lambda_0^m)$ is a torsion-free normal subgroup of Γ^+ that acts freely on $\mathcal{T}$. The groups $\Gamma^+(\lambda_0^{k+1})$, $k \geq 0$, are strictly decreasing, $|\Gamma^+(\lambda_0) : \Gamma^+(\lambda_0^{k+1})| = 11^{3k}$, and their intersection is trivial. Moreover $\Gamma^+(\lambda_0^m) \cong \{\gamma \in I[1/3]^1 : \gamma \equiv 1 \pmod{\lambda_0^m}\}$.

Proof (not formalized). The algebraic group B^1 of norm-one elements is simply connected and absolutely almost simple, and $B^1(E_3) = \mathrm{SL}_2(\mathbb{Q}_9)$ is not compact. Strong approximation [20, Section 7.4] makes $I[1/3]^1$ dense in $\prod_{\mathfrak{q}\nmid 3}(I \otimes \mathcal{O}_{E,\mathfrak{q}})^1$, so $I[1/3]^1 \to \mathrm{SL}_2(\mathcal{O}_E/\lambda_0^m)$ is onto. The kernel $\Gamma^+(\lambda_0^m)$ is normal. A nontrivial element of finite order fixes a vertex of $\mathcal{T}$ [23, Section I.4], so it lies in a Γ^+-conjugate of A_5 or of $\alpha A_5 \alpha^{-1}$; these map injectively to $\mathrm{PSL}_2(\mathbb{F}_{11})$, because reduction modulo λ_0 is injective on $2I$ ($1 + \lambda_0 M_2(\mathcal{O}_{E,\lambda_0})$ is torsion-free). So $\Gamma^+(\lambda_0^m)$ is torsion-free, and since Γ^+ acts without inversions, it acts freely. Since $-1 \not\equiv 1 \pmod{\lambda_0}$, each class $\pm\gamma$ in $\Gamma^+(\lambda_0^m)$ has exactly one representative $\gamma \equiv 1 \pmod{\lambda_0^m}$. The index is the order of $\ker(\mathrm{PSL}_2(\mathbb{Z}/11^{k+1}) \to \mathrm{PSL}_2(\mathbb{F}_{11}))$, which is 11^{3k}. If $g \in I[1/3]^1$ satisfies $g \equiv \pm 1 \pmod{\lambda_0^{k+1}}$ for every k, the sign is constant because $1 \not\equiv -1 \pmod{\lambda_0}$, so $g = \pm 1$ in $I \otimes \mathcal{O}_{E,\lambda_0}$, and $g = \pm 1$ because $B \to B \otimes E_{\lambda_0}$ is injective. □

Lemma 6.4 (Covering theory)**.** Not formalized; a complete proof is given below.

There is an isomorphism $\psi\colon F = \pi_1(X_0, \tau) \to \Gamma^+(\lambda_0)$ such that for every normal subgroup N of F of finite index the cover X_N of Lemma 3.6, with its lifted datum, is isomorphic to $\psi(N)\backslash\mathcal{T}$ with the datum descended from D_{tw}. Consequently $\dim_{\mathbb{F}_3} \ker \varphi_{F/N}(A) = 2 \dim_{\mathbb{F}_9} T_{\mathrm{tw}}(\psi(N)\backslash\mathcal{T})$.

Proof (not formalized). Since $\Gamma^+(\lambda_0)$ acts freely and without inversions, $\mathcal{T} \to X_0$ is a universal covering, and lifting the spanning tree τ to $\mathcal{T}$ gives an isomorphism ψ that sends the generator of an edge e to the deck transformation of the fundamental cycle of e. The identification must match

the data, not only the graphs, since X_0 has many automorphisms. We use the identification of Lemma 3.1, through which the data of Definition 3.4 are defined: labels are transported by coset representatives, so the datum of X_0 is the descent of D_{tw}. Then X_N and $\psi(N)\backslash\mathcal{T}$ carry the descent of the same datum. The last formula is Lemma 3.6. □

Let $N_k = \psi^{-1}(\Gamma^+(\lambda_0^{k+1}))$ for $k \geq 0$. By Lemma 6.3, (N_k) is a chain of F with $|F : N_k| = 11^{3k}$, and with $X_k = \Gamma^+(\lambda_0^{k+1})\backslash\mathcal{T}$, Lemma 6.4 gives

$$\dim_{\mathbb{F}_3} \ker \varphi_{F/N_k}(A) = 2 \dim_{\mathbb{F}_9} T_{\mathrm{tw}}(X_k). \tag{1}$$

6.2 Riemann–Roch on the Hilbert modular surface

Let $\mathfrak{n} = \lambda_0^{k+1}$, an ideal prime to 3 with a totally positive generator, and let $\mathcal{K}(\mathfrak{n}) = \ker\bigl(\mathrm{GL}_2(\hat{\mathcal{O}}_E) \to \mathrm{GL}_2(\mathcal{O}_E/\mathfrak{n})\bigr)$ be the principal congruence subgroup of level $\mathfrak{n}$. It is neat: a root of unity congruent to 1 modulo a prime above 11 has 11-power order, so it is 1 or has degree at least 10 over $\mathbb{Q}$, while the eigenvalues of elements of $\mathrm{GL}_2(E)$ have degree at most 4. Choose a smooth projective toroidal compactification, over $W(\bar{\mathbb{F}}_3)$, of the closure of one geometric connected component of the Hilbert modular surface of E of level $\mathcal{K}(\mathfrak{n})$, compatibly as k varies, and let S_k be its special fibre. The complex fibre of that component is $\Gamma(\mathfrak{n})\backslash\mathfrak{H}^2$ with $\Gamma(\mathfrak{n}) \subset \mathrm{SL}_2(\mathcal{O}_E)$, since E has narrow class number one and every totally positive unit is a square. We use the following published facts. The sources state them for full level $n \geq 3$, for $\Gamma_1(\mathfrak{c},\mathfrak{n})$-structures or for general neat levels; the facts we use are local on the moduli space or descend along the finite étale maps between levels prime to 3, so they hold at level $\mathcal{K}(\mathfrak{n})$.

(G1) S_k is smooth, projective and connected. Its boundary D_k is a disjoint union of $c_k \geq 1$ cycles of smooth rational curves, one for each cusp [21, Theorem 5.1, Proposition 5.2 and Corollary 5.3], [6], [8, Section 7], [18, Theorems 6.4.1.1 and 7.3.3.4].

(G2) Since 3 is unramified in E, $\mathcal{O}_E \otimes W = W \times W$ and the Hodge bundle splits as $\omega_1 \oplus \omega_2$. The Kodaira–Spencer map with logarithmic poles gives $\Omega^1_{S_k}(\log D_k) \cong \omega_1^2 \oplus \omega_2^2$ [18, Theorem 6.4.1.1(4)]: Lan's Kodaira–Spencer sheaf is the symmetric square of ω over $\mathcal{O}_E \otimes \mathcal{O}$, because ω is locally free of rank one over $\mathcal{O}_E \otimes \mathcal{O}$ and a polarization of degree prime to 3 identifies the Hodge bundle of the dual with ω. Hence

$$K_{S_k} + D_k \cong \omega_1^2 \otimes \omega_2^2 =: L,$$

the bundle of parallel weight 2.

(G3) The partial Hasse invariants $h_1 \in H^0(\omega_1^{-1}\omega_2^3)$ and $h_2 \in H^0(\omega_1^3\omega_2^{-1})$ have reduced divisors Z_1, Z_2, the Goren–Oort divisors, which are regular curves meeting in finitely many points [9, Section 1 and Theorem 2.1(1)]; see also [10, 26]. They extend to the compactification by the Koecher principle [8, Theorem 8.3], and they do not vanish on D_k, because their q-expansion at every cusp is 1 [9, Theorem 2.1(2)]. So $h_1h_2 \in H^0(L)$ has divisor $Y_k = Z_1 + Z_2$ and $L(-Y_k) \cong \mathcal{O}$. Since 3 is inert, Y_k is the reduced supersingular locus [1, Proposition 4.1].

(G4) $\chi(\mathcal{O}_{S_k}) = V_k := e_{\mathrm{top}}/4 = 11^{3k+1}$, where e_{top} is the Euler number of $\Gamma(\mathfrak{n})\backslash\mathfrak{H}^2$. Since $\chi(\mathcal{O})$ is constant in flat families and a birational invariant, it may be computed in characteristic 0 with any resolution of the cusps. Every cusp of $\Gamma(\mathfrak{n})$ is conjugate to ∞ and has type $(\mathfrak{n},\mathcal{U}_\mathfrak{n})$, which is isomorphic to $(\mathcal{O}_E,\mathcal{U}_\mathfrak{n})$ through a totally positive generator of $\mathfrak{n}$, where $\mathcal{U}_\mathfrak{n} = \{\varepsilon^2 : \varepsilon \in \mathcal{O}_E^\times,\ \varepsilon \equiv 1 \pmod{\mathfrak{n}}\}$. The module $\mathcal{O}_E = \mathbb{Z} + \mathbb{Z}\theta^2$ has reduced number $\theta^2 = 3 - 1/\theta^2$, with purely periodic minus continued fraction $[[3]]$, so every cusp is resolved by

a cycle of $[\mathcal{O}_E^{\times,+} : \mathcal{U}_\mathfrak{n}]$ curves of self-intersection -3 [11, Sections 2.5 and 2.6], [27]: 5 curves for $k = 0$, since $\theta \equiv 4$ has order 5 modulo λ_0, and 55 for $k = 1$, since $\theta \equiv 37$ has order 55 modulo λ_0^2. Noether's formula, with $K^2 = 2e_{\mathrm{top}} + \sum_j (2 - b_j)$ and $c_2 = e_{\mathrm{top}} + \#\{\text{curves}\}$ for curves of self-intersection $-b_j = -3$, gives $\chi = (K^2 + c_2)/12 = e_{\mathrm{top}}/4 + \sum_j (3 - b_j)/12 = e_{\mathrm{top}}/4$. Finally $e_{\mathrm{top}} = |\mathrm{PSL}_2(\mathbb{F}_{11})| \cdot 2\zeta_E(-1) = 660/15 = 44$ at level λ_0, since $2\zeta_E(-1) = 1/15$ [11, Sections 1.6 and 1.7], and e_{top} is multiplied by 11^{3k} at level λ_0^{k+1}.

Lemma 6.5 (Sections on the Goren–Oort divisor)**.** Not formalized; a complete proof is given below. $h^0(Y_k, L|_{Y_k}) \geq V_k - 1$.

Proof (not formalized). By Riemann–Roch, $\chi(L) = \chi(\mathcal{O}) + \frac{1}{2} L \cdot (L - K) = \chi(\mathcal{O}) + \frac{1}{2}(K + D) \cdot D$. Each boundary cycle D_j has arithmetic genus 1 and the D_j are disjoint, so adjunction gives $(K + D) \cdot D = \sum_j (K + D_j) \cdot D_j = 0$. Hence $\chi(L) = \chi(\mathcal{O}) = V_k$ by (G4). By Serre duality $h^2(L) = h^0(K - L) = h^0(-D) = 0$, because $D \neq 0$ and S_k is integral. So $h^0(L) \geq \chi(L) = V_k$. The exact sequence $0 \to L(-Y_k) \cong \mathcal{O} \to L \to L|_{Y_k} \to 0$ and $h^0(\mathcal{O}) = 1$ give $h^0(Y_k, L|_{Y_k}) \geq h^0(L) - 1 \geq V_k - 1$. □

If moreover $h^1(\mathcal{O}_{S_k}) = 0$, then $h^1(L) = h^1(-D) = c_k - 1$ and $h^0(L|_{Y_k}) = V_k + c_k - 2$ exactly; the proof uses only the lower bound.

6.3 Uniformization of the supersingular locus

We use the description of the supersingular locus of Hilbert–Blumenthal surfaces at an inert prime by Stamm, as presented by Kudla and Rapoport [17, Section 4], [24], the uniformization theorem of Rapoport and Zink [22, Theorem 6.30], and the intersection behaviour of the Goren–Oort divisors found by Bachmat and Goren [1, Theorem 6.1]. Kudla and Rapoport work with abelian varieties of dimension 8 with an action of $C(V) \otimes k$, for the Clifford algebra $C(V)$ of a quadratic space attached to a quaternion algebra B_0 over $\mathbb{Q}$; for $B_0 = M_2(\mathbb{Q})$ and $k = E$ their moduli space is the Hilbert–Blumenthal surface, and Morita equivalence identifies their Dieudonné lattices with ours [17, Introduction and Section 3].

The local picture. Let $\mathcal{M}$ be the Rapoport–Zink space of supersingular polarized 3-divisible groups with $\mathcal{O}_E \otimes \mathbb{Z}_3$-action, with framing object $\mathbb{X}$: a point is such a group X with a framing quasi-isogeny ϱ between X and $\mathbb{X}$. The group J of $\mathcal{O}_E$-linear self-quasi-isogenies of $\mathbb{X}$ that multiply the polarization by a factor in $\mathbb{Q}_3^\times$ acts on $\mathcal{M}$ by changing ϱ, so the Hodge bundles ω_1, ω_2 of the universal 3-divisible group, and $L = \omega_1^2 \otimes \omega_2^2$, carry a canonical J-action. Following [17, Section 4], the isocrystal of $\mathbb{X}$ yields two-dimensional $\mathbb{Q}_9$-spaces U_0 and U_1 with alternating forms $\langle\,,\rangle$, on which J acts, and Frobenius gives a σ-semilinear bijection $\Phi\colon U_1 \to U_0$ that commutes with J; Frobenius maps U_0 to U_1 by $3\Phi^{-1}$, and $\langle \Phi x, \Phi y\rangle = 3\sigma\langle x, y\rangle$. For $n \in \mathbb{Z}$ let $\mathcal{M}_{\mathrm{red}}^{(n)}$ be the open and closed part of the reduced locus whose Dieudonné lattices $L_0 \oplus L_1$ satisfy $\langle L_i, L_i\rangle = 3^n W$; its stabilizer J^0 in J consists of the elements whose factor is a 3-adic unit. By [17, Lemma 4.3 and Proposition 4.4], $\mathcal{M}_{\mathrm{red}}^{(n)}$ is a union of projective lines over $\bar{\mathbb{F}}_3$: a line $C_\Lambda = \mathbb{P}^1(\Lambda/3\Lambda)$ of the first kind for each $\mathbb{Z}_9$-lattice $\Lambda \subset U_0$ with $\langle \Lambda, \Lambda\rangle = 3^n\mathbb{Z}_9$, and a line $C_{\Lambda'} = \mathbb{P}^1(\Lambda'/3\Lambda')$ of the second kind for each $\mathbb{Z}_9$-lattice $\Lambda' \subset U_1$ with $\langle \Lambda', \Lambda'\rangle = 3^n\mathbb{Z}_9$. Lines of the same kind are disjoint. The lines C_Λ and $C_{\Lambda'}$ meet if and only if $3\Lambda \subset \Phi\Lambda' \subset \Lambda$, and then in one point, which is $\Phi\Lambda'/3\Lambda$ on C_Λ and $\Phi^{-1}(3\Lambda)/3\Lambda'$ on $C_{\Lambda'}$ [17, Lemma 4.3(ii)]. Each line meets exactly one line of the other kind at each of its ten $\mathbb{F}_9$-rational points, the superspecial points, and no line elsewhere [17, Lemma 4.3(iii) and the text after (4.9)]. As schemes over $\mathbb{F}_9$, the lines are the projective lines $\mathbb{P}^1(\Lambda/3\Lambda)$ and $\mathbb{P}^1(\Lambda'/3\Lambda')$, with quasi-isogenies acting through the lattices: the morphism from their union to the supersingular locus

is its normalization, equivariantly [17, (4.10)], after Stamm [24]. The crossings are ordinary double points [1, Theorem 6.1]. These two statements concern the global moduli space, and they transfer to $\mathcal{M}$ through the uniformization theorem used below; the proof of the second is local deformation theory. Scaling a lattice by 3 multiplies $\langle\,,\rangle$ by 9, and $\langle\Phi\Lambda', \Phi\Lambda'\rangle = 3^{n+1}\mathbb{Z}_9$; so $C_\Lambda \mapsto [\Lambda]$ and $C_{\Lambda'} \mapsto [\Phi\Lambda']$ map the lines of the two kinds bijectively to the homothety classes of lattices in U_0 of the two parities, and they identify the dual graph of $\mathcal{M}^{(n)}_{\mathrm{red}}$ with the tree $\mathcal{T}_{U_0}$ of U_0, equivariantly for J^0.

Lemma 6.6 (Uniformization)**.** Not formalized; a complete proof is given below.

There exist an integer n, an action of Γ^+ on $\mathcal{M}^{(n)}_{\mathrm{red}}$ through J^0, and an isomorphism $U_0 \cong \mathbb{Q}_9^2$ that carries $\mathcal{T}_{U_0}$ to $\mathcal{T}$, with the following properties.

(a) The resulting identification of the dual graph of $\mathcal{M}^{(n)}_{\mathrm{red}}$ with $\mathcal{T}$ is equivariant for the action of Γ^+ on $\mathcal{T}$ of Section 3.

(b) For every $k \geq 0$, $Y_k \cong \Gamma^+(\lambda_0^{k+1})\backslash\mathcal{M}^{(n)}_{\mathrm{red}}$, and under this isomorphism $L|_{Y_k}$ is the descent of L with its J-action. In particular the dual graph of Y_k is X_k.

(c) L has degree 4 on every line of $\mathcal{M}^{(n)}_{\mathrm{red}}$.

Proof (not formalized). The global picture. Let $G' = \{g \in B^\times : \operatorname{Nrd} g \in \mathbb{Q}^\times\}$, write $\mathcal{K}(\mathfrak{n}) = \mathcal{K}_3\mathcal{K}^{(3)}$ with $\mathcal{K}^{(3)}$ the part away from 3, and let $\mathbb{A}_f^{(3)}$ be the finite adeles away from 3. Fix a supersingular base point x_0 of S_0 with integral level structure, which exists by Lemma 6.5, with compatible lifts to the S_k, and frame $\mathcal{M}$ by its 3-divisible group. The supersingular points form one isogeny class, the class I_0 of [17, Section 3], which is basic. The uniformization theorem [22, Theorem 6.30] identifies the formal completion of the Shimura variety of level $\mathcal{K}(\mathfrak{n})$ along an open and closed part Z' of the supersingular locus Z with $G'(\mathbb{Q})\backslash(\mathcal{M} \times G'(\mathbb{A}_f^{(3)})/\mathcal{K}^{(3)})$, a finite disjoint union of quotients of $\mathcal{M}$. Here $G'(\mathbb{Q})$ is the group of $\mathcal{O}_E$-linear self-quasi-isogenies of x_0 that multiply the polarization by a factor in $\mathbb{Q}^\times$; it acts on $\mathcal{M}$ through J, and on $G'(\mathbb{A}_f^{(3)})$ through the Tate modules of x_0 and its level structure. Moreover $Z' = Z$, because the group $G^* = \{g \in \mathrm{GL}_2(E) : \det g \in \mathbb{Q}^\times\}$ of the moduli problem satisfies the Hasse principle: its derived group $\operatorname{Res}_{E/\mathbb{Q}}\mathrm{SL}_2$ is simply connected, so it satisfies the Hasse principle [20, Chapter 6]; $G^*/G^{*,\mathrm{der}} \cong \mathbb{G}_m$ has trivial first Galois cohomology; and $\det\colon G^*(\mathbb{Q}_v) \to \mathbb{Q}_v^\times$ is onto at every place v. The endomorphisms of x_0, with 3 inverted, form a maximal $\mathcal{O}_E[1/3]$-order of the definite quaternion algebra B, by Tate's theorem [25] at the primes $\ell \neq 3$. Since B has class number one, it has type number one [28, Corollaire I.4.11 and p. 155], and we may choose the identification of B with the endomorphism algebra of x_0 so that this order is $I[1/3]$. By the norm theorem of Hasse and Schilling, which gives $\mathbb{Q}_{>0} \subseteq \operatorname{Nrd} B^\times$ [28, Theorem III.4.1], and strong approximation for B^1 with respect to $\{3\}$, every supersingular point of S_k has a representative in the slice $\mathcal{M}^{(n)}_{\mathrm{red}} \times \{\mathcal{K}^{(3)}\}$ that contains x_0. The stabilizer of that slice is $\{\gamma \in G'(\mathbb{Q}) \cap \mathcal{K}^{(3)} : v_3(\operatorname{Nrd}\gamma) = 0\}$. Its reduced norms are positive rationals that are units at every prime, so they equal 1, and the stabilizer is $\{\gamma \in I[1/3]^1 : \gamma \equiv 1 \pmod{\mathfrak{n}}\} \cong \Gamma^+(\lambda_0^{k+1})$ (Lemma 6.3). The element $-1 \in I[1/3]^1$ acts trivially on $\mathcal{M}_{\mathrm{red}}$, since (X, ϱ) is isomorphic to $(X, -\varrho)$ through $-1 \in \operatorname{Aut} X$, and on L, by $(-1)^4 = 1$; so $I[1/3]^1$ acts through Γ^+, and through J^0 since its reduced norms are 1. The uniformization is compatible with the universal 3-divisible groups, hence with ω_1, ω_2 and L. This proves (b), given (a).

The identification with $\mathcal{T}$. The algebra B acts on the isocrystal of x_0 contravariantly. Composed with the main involution $b \mapsto \bar{b}$, this action gives a homomorphism $B \otimes_E E_3 \to \mathrm{End}_{\mathbb{Q}_9}(U_0)$, an isomorphism of central simple algebras, which on elements of reduced norm one is their action on

lattices. Here E_3 is identified with $\mathbb{Q}_9$ through the embedding of $\mathcal{O}_E \otimes \mathbb{Z}_3$ into W that defines L_0. By the Skolem–Noether theorem we may choose $U_0 \cong \mathbb{Q}_9^2$ so that this homomorphism becomes the isomorphism $B \otimes_E E_3 \cong M_2(\mathbb{Q}_9)$ of Section 3, composed with σ if that embedding reduces to $\theta \mapsto 2 - i$ (a case treated in Lemma 6.9); then the identification of the dual graph with $\mathcal{T}_{U_0} \cong \mathcal{T}$ is Γ^+-equivariant, which is (a).

The degree. Since $\Gamma^+(\lambda_0^{k+1})$ acts freely on $\mathcal{T}$ (Lemma 6.3), every line maps isomorphically onto a component of Y_k, compatibly with L. Let C be a component of Z_1, and put $a = \deg \omega_1|_C$ and $b = \deg \omega_2|_C$. The section h_2 vanishes on C exactly at the ten superspecial points, simply, because the crossings are ordinary double points of the reduced curve $Z_1 + Z_2$ [1, Theorem 6.1]; so $3a - b = 10$. Lines of the same kind are disjoint, so $C^2 = Z_1 \cdot C = -a + 3b$, and adjunction with $K = L - D$ and $D \cdot C = 0$ gives $-2 = (K + C) \cdot C = a + 5b$. Hence $(a, b) = (3, -1)$ and $\deg L|_C = 2a + 2b = 4$; the components of Z_2 are symmetric. This is (c). As a check, $Z_1 \sim 3\omega_2 - \omega_1$, $Z_2 \sim 3\omega_1 - \omega_2$ and $\omega_i^2 = 0$ give $Z_1 \cdot Z_2 = 10\,\omega_1\omega_2 = 10\,e_{\mathrm{top}}/4$, while each of the n_1 lines of one kind meets ten lines of the other, so $Z_1 \cdot Z_2 = 10n_1$; thus $n_1 = e_{\mathrm{top}}/4 = 11^{3k+1} = |\Gamma^+ : \Gamma^+(\lambda_0^{k+1})|/60$, as X_k requires. □

Geometric data. A *geometric datum* on $\mathcal{T}$ is a Γ^+-equivariant family consisting of a smooth rational curve C_v over $\bar{\mathbb{F}}_3$ for each vertex v, on which the stabilizer of v acts faithfully; a line bundle L_v of degree 4 on C_v, with an action of that stabilizer; for each edge e at v a point $P_{v,e} \in C_v$, the *special points*; and for each edge $e = \{v, w\}$ an isomorphism between the fibres of L_v at $P_{v,e}$ and of L_w at $P_{w,e}$. Its *type* is the pair of the isomorphism classes of the actions of the stabilizers of Λ_0 and Λ_1 on C_{Λ_0} and C_{Λ_1}. For a subgroup $\Gamma \subseteq \Gamma^+(\lambda_0)$ of finite index, its *Tanner code* on $\Gamma\backslash\mathcal{T}$ is the space of Γ-invariant families of sections $s_v \in H^0(C_v, L_v)$ whose values at the two special points of each edge correspond.

By Lemma 6.6, the lines of $\mathcal{M}_{\mathrm{red}}^{(n)}$ with the restrictions of L, their superspecial points and the identifications of the fibres of L at the crossings form a geometric datum D_{geo} on $\mathcal{T}$, the same for every k. The stabilizers act faithfully on the lines: the stabilizer of a vertex acts on its line through its action on $\Lambda/3\Lambda$, or, for a line of the second kind, through its action on $\Phi\Lambda'/3\Phi\Lambda'$ conjugated by σ, since J commutes with Φ; and $2I \to \mathrm{SL}_2(\mathbb{F}_9)$ is injective (proof of Lemma 6.2).

Lemma 6.7 (Sections are Tanner codes)**.** Not formalized; a complete proof is given below.

For every subgroup $\Gamma \subseteq \Gamma^+(\lambda_0)$ of finite index, $H^0(\Gamma\backslash\mathcal{M}_{\mathrm{red}}^{(n)}, L)$ is isomorphic to the Tanner code of D_{geo} on $\Gamma\backslash\mathcal{T}$.

Proof (not formalized). Since Γ acts freely on $\mathcal{T}$ (Lemma 6.3), the curve $Y = \Gamma\backslash\mathcal{M}_{\mathrm{red}}^{(n)}$ has the lines as components and ordinary nodes exactly at the superspecial points. So $0 \to \mathcal{O}_Y \to \nu_*\mathcal{O}_{\tilde{Y}} \to \bigoplus_{\mathrm{nodes}} \bar{\mathbb{F}}_3 \to 0$ for the normalization ν, and a section of L on Y is a family of sections on the components whose values at each node correspond, that is, an element of the Tanner code. □

6.4 Rigidity and the twist

Let $\rho\colon 2I \to \mathrm{SL}_2(\mathbb{F}_9)$ be reduction modulo 3 through $I \otimes_{\mathcal{O}_E} \mathcal{O}_{E,3} \cong M_2(\mathbb{Z}_9)$, that is, the action of $2I$ on $\Lambda_0/3\Lambda_0 = \mathbb{F}_9^2$. It is one of the two representations of Lemma 6.1(a), and the other is ρ^σ, obtained by applying σ to the entries. Applying σ to the coefficients turns a datum into another one, whose Tanner codes are the images of the old ones under σ, so of the same dimension.

The data D_{untw} and D_{tw} of Section 3.2 are geometric data: at every vertex take $\mathbb{P}^1$ with $\mathcal{O}(4)$, whose sections are the binary quartic forms; take as special points the points $[x]$ of the labels x; and at an edge with labels x and y at its two ends, identify the fibres so that sections f and g correspond when $f(x) = g(y)$, which is well defined because labels are defined up to scalars c with $c^4 = 1$. Since

evaluation of quartics at ten points is injective (Lemma 3.3), the Tanner code of D_{tw} on $\Gamma\backslash\mathcal{T}$ in this sense is $T_{\mathrm{tw}}(\Gamma\backslash\mathcal{T}) \otimes_{\mathbb{F}_9} \bar{\mathbb{F}}_3$. In both data the stabilizer of Λ_0 acts through ρ, since $u \in 2I$ maps the edge $u'e_0$, with label $u'p_0$, to the edge $uu'e_0$, with label $\rho(u)u'p_0$. In the coordinates transported by α, the stabilizer of Λ_1 acts through $\rho \circ \operatorname{Ad}\alpha^{-1}$ in D_{untw}, and through $(\rho \circ \operatorname{Ad}\alpha^{-1})^\sigma$ in D_{tw}, since $s = \alpha s'\alpha^{-1}$ maps the label $\mathrm{Frob}(t'k_0)$ to $\mathrm{Frob}(s't'k_0) = \mathrm{Frob}(s')\,\mathrm{Frob}(t'k_0)$. So D_{untw} has type $(\rho, \rho \circ \operatorname{Ad}\alpha^{-1})$ and D_{tw} has type $(\rho, (\rho \circ \operatorname{Ad}\alpha^{-1})^\sigma)$.

Lemma 6.8 (Rigidity)**.** Not formalized; a complete proof is given below.

Let D and D' be geometric data on $\mathcal{T}$.

(a) If D and D' have the same type, their Tanner codes on $\Gamma\backslash\mathcal{T}$ are isomorphic for every subgroup $\Gamma \subseteq \Gamma^+(\lambda_0)$ of finite index.

(b) Every geometric datum has the type of one of D_{untw}, D_{tw}, D^σ_{untw} and D^σ_{tw}, and these four types are different.

Proof (not formalized). (1) By Lemma 6.1(a), the stabilizer A_5 of Λ_0 acts on C_{Λ_0} through ρ or ρ^σ, up to isomorphism, and $\alpha A_5\alpha^{-1}$ acts on C_{Λ_1} through $\rho \circ \operatorname{Ad}\alpha^{-1}$ or its σ-twist. So there are at most four types, and since $\rho \not\cong \rho^\sigma$, the four data in (b) have four different ones. This is (b).

(2) By Lemma 6.1(b), an equivariant isomorphism between two such vertex curves is unique. An A_5-linearization of $\mathcal{O}(4)$ is unique because A_5 is perfect, and its equivariant automorphisms are scalars.

(3) The ten special points at a vertex are permuted by its stabilizer, transitively with stabilizer S_3 (Lemma 6.2), so by Lemma 6.1(c) they form the orbit of size 10; since S_3 is self-normalizing, the equivariant bijection between the edges at a vertex and its special points is unique.

(4) Take D and D' of the same type, fix equivariant isomorphisms of their vertex data at Λ_0 and Λ_1, unique up to scalars by (2) and (3), and transport them by Γ^+. The edge stabilizers act on the fibres at the special points through the same character, the inverse fourth power of the character of Γ_e on the line of the edge, which is trivial (Section 3.2). So the ratio of the two identifications at an edge is a Γ^+-invariant function on edges, hence a constant r by edge-transitivity (Lemma 6.2), and scaling the sections at the odd vertices by r is an isomorphism of Tanner codes on every $\Gamma\backslash\mathcal{T}$. This is (a). □

Lemma 6.9 (The twisted type)**.** Not formalized; a complete proof is given below.

D_{geo} has the type of D_{tw} or of D^σ_{tw}. Hence $h^0(Y_k, L|_{Y_k}) = \dim_{\mathbb{F}_9} T_{\mathrm{tw}}(X_k)$ for every $k \geq 0$.

Proof (not formalized). We use the local picture of Section 6.3 and the identification $U_0 \cong \mathbb{Q}_9^2$ of Lemma 6.6.

(1) An element $g \in J$ commutes with Φ. So if b_1, b_2 is a basis of U_0 and U_1 is given the basis $\Phi^{-1}b_1, \Phi^{-1}b_2$, the matrix of g on U_1 is $\sigma^{-1} = \sigma$ applied to its matrix on U_0.

(2) Hence the stabilizer of a vertex acts on a line C_Λ of the first kind through its action on $\Lambda/3\Lambda$, and on a line $C_{\Lambda'}$ of the second kind through its action on $M/3M$, for $M = \Phi\Lambda'$, with the entries conjugated by σ: Φ identifies $C_{\Lambda'}$ with the Frobenius twist of $\mathbb{P}^1(M/3M)$.

(3) Suppose that the lines at the even vertices are of the first kind. Then A_5 acts on C_{Λ_0} through ρ, and $\alpha A_5\alpha^{-1}$ acts on C_{Λ_1} through its action on $\alpha\Lambda_0/3\alpha\Lambda_0$ conjugated by σ, which in the coordinates transported by α is $(\rho \circ \operatorname{Ad}\alpha^{-1})^\sigma$. This is the type of D_{tw}. The special points match the labels of Section 3.2. Up to the same power of 3, the lattices at Λ_0 and Λ_1 are Λ_0 and $M = \alpha\Lambda_0$. On C_{Λ_0} the special point towards Λ_1 is $\alpha\Lambda_0/3\Lambda_0 = \mathrm{im}(\alpha \bmod 3) = \langle p_0\rangle$. On C_{Λ_1} the special point towards Λ_0 is $\Phi^{-1}(3\Lambda_0)/3\Lambda'$, which Φ maps to $3\Lambda_0/3\alpha\Lambda_0$; in the coordinates transported by α this is $\bar\alpha\Lambda_0/3\Lambda_0 = \ker(\alpha \bmod 3) = \langle k_0\rangle$, so the special point is $[\mathrm{Frob}(k_0)]$.

(4) If the lines at the even vertices are of the second kind, the same computation gives the type $(\rho^\sigma, \rho \circ \operatorname{Ad} \alpha^{-1})$, which is that of D_{tw}^σ.

(5) If the embedding of $\mathcal{O}_E \otimes \mathbb{Z}_3$ into W that defines L_0 reduces to $\theta \mapsto 2 - i$, both entries of the type are replaced by their images under σ. In every case the type is that of D_{tw} or of D_{tw}^σ.

For the last claim, $Y_k \cong \Gamma^+(\lambda_0^{k+1}) \backslash \mathcal{M}_{\mathrm{red}}^{(n)}$ (Lemma 6.6), so by Lemmas 6.7 and 6.8 $h^0(Y_k, L|_{Y_k})$ is the dimension over $\bar{\mathbb{F}}_3$ of the Tanner code on X_k of D_{tw} or of D_{tw}^σ. This is $\dim_{\mathbb{F}_9} T_{\mathrm{tw}}(X_k)$: the first code is $T_{\mathrm{tw}}(X_k) \otimes_{\mathbb{F}_9} \bar{\mathbb{F}}_3$, and the second its image under σ. □

6.5 Conclusion

Theorem 6.10 (Theorem H)**.** Not formalized; a complete proof is given below.
For the chain $N_k = \psi^{-1}(\Gamma^+(\lambda_0^{k+1}))$ of F, with $|F : N_k| = 11^{3k}$, and every $k \geq 0$,

$$\dim_{\mathbb{F}_3} \ker \varphi_{F/N_k}(A) \;\geq\; 22\,|F : N_k| - 2 \;\geq\; 20\,|F : N_k|.$$

In particular hypothesis (KD) holds with $\varepsilon = 20$, and so does (KN).

Proof (not formalized). By Eq. (1), Lemma 6.9 and Lemma 6.5,

$$\dim_{\mathbb{F}_3} \ker \varphi_{F/N_k}(A) = 2 \dim_{\mathbb{F}_9} T_{\mathrm{tw}}(X_k) = 2h^0(Y_k, L|_{Y_k}) \geq 2(V_k - 1) = 22\,|F : N_k| - 2,$$

since $V_k = 11^{3k+1} = 11\,|F : N_k|$; and $22|F : N_k| - 2 \geq 20|F : N_k|$ because $|F : N_k| \geq 1$. □

At $k = 0$ the last inequality is an equality, $20 = 22 - 2$, while the actual kernel dimension is 42 (Remark 4.2).

6.6 Numerical checks

Remark 6.11 (Numerical)**.** The script `tools/lambda_level1.py` computes the first nested level $k = 1$ exactly. It lifts the coset representatives to $\mathrm{PSL}_2(\mathbb{Z}/121)$, obtains voltages in $\ker(\mathrm{PSL}_2(\mathbb{Z}/121) \to \mathrm{PSL}_2(\mathbb{F}_{11})) \cong \mathbb{F}_{11}^3$, and sums the kernels of the 1331 twisted 55×55 operators over $\mathbb{F}_{3^{10}}$. With c_k the number of cusps ($c_0 = 12$, $c_1 = 132$):

k	$\lvert F : N_k \rvert$	V_k	c_k	$\dim T_{\mathrm{tw}}(X_k)$	$V_k + c_k - 2$	$V_k - 1$	$\dim T_{\mathrm{untw}}(X_k)$
0	1	11	12	21	21	10	20
1	1331	14641	132	14771	14771	14640	2460

So the exact formula $V_k + c_k - 2$ holds at $k = 0, 1$. At $k = 1$ the untwisted datum falls below the bound of Lemma 6.5, which confirms Lemma 6.9 independently, by elimination between the types of Lemma 6.8. By Lemmas 3.6 and 6.4 the value 14771 means $\dim_{\mathbb{F}_3} \ker \varphi_{F/N_1}(A) = 29542$, a normalized kernel of about 22.2. Each of the 1330 nontrivial characters of $\mathbb{F}_{11}^3$ contributes a twisted kernel of dimension 11 or 12. With random voltages in $\mathbb{F}_{11}^3$ on the same graph, which define connected covers of the same degree that are not congruence covers (Remark 7.3), only the kernel of dimension 21 pulled back from X_0 survives (`tools/lambda_level1.pyrandoms`, $s = 1, 2, 3$). The finite group theory of Lemma 6.1 is also checked by exhaustive computation in `tools/a5_checks.py`.

7 What a proof of the hypothesis needs

Theorem A is formalized except for hypothesis (KN). This section explains why (KN), and any statement that could replace it, needs an argument about infinitely many levels, and why for the matrix A that argument cannot apply to all covers of the base graph; heuristically, it must use the arithmetic of congruence covers. Nothing in this section is used in a proof.

Remark 7.1 (Limits ignore finitely many levels)**.** Part (2) of the conjecture is a statement about limits, and changing a rank sequence at finitely many places changes no limit. So no finite computation, however large, can refute part (2): a refutation must prove something about infinitely many levels of some chain. The proof of Theorem P does this with a finite certificate (Lemma 4.1) and two uniform arguments, the induction over the coordinates u^k in Lemma 4.3 and the monotonicity of Lemma 4.4. For the congruence chain the kernel is certified at $k = 0$ (Remark 4.2) and computed at $k = 1$ (Remark 6.11), but no uniform step that carries them to all levels is known apart from Section 6.

Remark 7.2 (No monotonicity along ℓ-steps)**.** For a prime $\ell \neq 3$, call $N' \leq N$ an ℓ-step if N/N' is an ℓ-group. The steps of the congruence chain are 11-steps, and along ℓ-steps, in characteristic 3, the normalized kernel can go up or down. Let x be a free generator of F and consider the cyclic 11-cover $F \to \mathbb{Z}/11$ that sends x to a generator. Up: the 1×1 matrix $a = 1 + x + \cdots + x^{10}$ over $\mathbb{F}_3[F]$ has normalized kernel 0 at F itself, where $a \mapsto 11 \neq 0$, and $10/11$ on the cyclic 11-cover, where a maps to the norm element, whose kernel is the augmentation ideal. Down: the 1×1 matrix $1 - x$ has normalized kernel 1 at F and $1/11$ on the cyclic 11-cover. So, unlike Lemma 4.4, no general inequality passes a bound from one level of the congruence chain to the next. (Remark 6.11 records the values for A itself at $k = 0, 1$.)

Remark 7.3 (Congruence separates the chains)**.** Proof sketch only: no proper normal subgroup of F of 3-power index is a congruence subgroup, that is, contains some $\psi^{-1}(\Gamma^+(\mathfrak{m}) \cap \Gamma^+(\lambda_0))$, where for an ideal $\mathfrak{m}$ prime to 3 we write $\Gamma^+(\mathfrak{m})$ for the image in Γ^+ of $\{\gamma \in I[1/3]^1 : \gamma \equiv 1 \pmod{\mathfrak{m}}\}$. Indeed, by strong approximation the closure of (a lift to $I[1/3]^1$ of) $\Gamma^+(\lambda_0)$ in $\prod_{\mathfrak{q} \nmid 3} \mathrm{SL}_2(\mathcal{O}_{E,\mathfrak{q}})$ is $\mathcal{P} \times \prod_{\mathfrak{q} \nmid 3\lambda_0} \mathrm{SL}_2(\mathcal{O}_{E,\mathfrak{q}})$, with $\mathcal{P} = \ker(\mathrm{SL}_2(\mathbb{Z}_{11}) \to \mathrm{SL}_2(\mathbb{F}_{11}))$ a pro-11 group. A finite 3-group quotient of this product is trivial on $\mathcal{P}$ and on the pro-p congruence kernel of each other factor ($p \neq 3$ the residue characteristic), and what remains is a product of groups $\mathrm{SL}_2(\mathbb{F}_q)$ with $q \geq 4$, since the residue fields of E away from 3 are $\mathbb{F}_4, \mathbb{F}_5, \mathbb{F}_{11}, \ldots$; these groups are perfect. The same argument shows that N_1 is the only congruence subgroup of F with quotient $\mathbb{F}_{11}^3$, because the Frattini quotient of $\mathcal{P}$ is $\mathfrak{sl}_2(\mathbb{F}_{11}) \cong \mathbb{F}_{11}^3$.

So the random $\mathbb{F}_{11}^3$-covers of Remark 6.11 are not congruence covers, and every proper level of a pro-3-cofinal chain with 3-group quotients is a non-congruence subgroup; along such chains the normalized kernel tends to 0 (Theorem P). Independently of this sketch, Theorem P shows that no proof of (KN) for the matrix A can apply to every cover of X_0. Nor can it work in characteristic 0: for an integral lift of A the ranks over $\mathbb{Q}$ dominate those over $\mathbb{F}_3$, so they tend to 110 along the chains of Theorem P, hence, by Lück's theorem, along every chain. Heuristically, the property that distinguishes the congruence covers is their moduli interpretation, as supersingular loci of Hilbert modular surfaces with level structure, and that is what Section 6 uses.

Remark 7.4 (Twists only move the problem)**.** A step $N_k \to N_{k+1}$ has abelian quotient $\mathbb{F}_{11}^3$, of order prime to 3, so after extending scalars to contain the 11-th roots of unity, the kernel at level $k+1$ is the sum over the 11^3 characters ξ of N_k/N_{k+1} of the kernels of ξ-twisted operators on the level-k cover. Keeping the normalized kernel at ε requires the twisted kernels to have normalized

dimension ε on average, a statement of the same kind about forms with a character. Remark 6.11 records the twisted kernels at $k = 1$.

Remark 7.5 (The shape of a counterexample)**.** Over a free group every limit of a rank sequence is at most the inner rank (Section 5.1). A counterexample to part (2) consists of two chains with convergent rank sequences and limits $l_1 > l_2$, so along the second chain the ranks are at most $l_1 - \varepsilon$ infinitely often. For a square matrix whose inner rank is reached along some chain, this is the shape of (KN), and Remark 7.1 applies. A counterexample of another shape would still need an argument about infinitely many levels.

What a complete formalization needs. Formalizing Section 6 needs Hilbert modular surfaces and their toroidal compactifications over $\mathbb{Z}_{(3)}$, coherent cohomology with Serre duality and Riemann–Roch for surfaces, the logarithmic Kodaira–Spencer isomorphism, partial Hasse invariants, and the uniformization of the supersingular locus. None of this is in Mathlib. Two smaller steps would move parts of the informal argument into Lean.

- *The explicit chain.* The groups N_k can be defined in Lean by 89 explicit matrices modulo 11^{k+1}, the images of the generators. Strictness follows from the binomial theorem: if $g \equiv 1$ modulo 11^k but not modulo 11^{k+1}, then the same holds for g^{11} one level deeper. Trivial intersection follows from the injectivity of $F \to \Gamma^+(\lambda_0)$, which a ping-pong argument on the ends of $\mathcal{T}$ would prove (Mathlib has the ping-pong lemma for free groups). Hypothesis (KD) would then concern an explicit chain.
- *More levels.* A direct certificate for the kernel at $k = 1$ would have about $4 \cdot 10^9$ entries (29542 vectors of length 146410), beyond the reach of the kernel checker. Through the character decomposition of Remark 7.4 it would be 1331 small certificates over $\mathbb{F}_{3^5}$ and a formal Fourier decomposition. This would add one level and would not change the logical status of Theorem A.

8 Open problems

The arbitrary-field Lück approximation conjecture is refuted by Theorem A, which is formal except for Theorem H. The following questions remain.

1. *A formal or elementary proof of Theorem H.* Formalize the geometry of Section 6, or find another proof of (KN), for this or any square matrix of full inner rank over a free group in positive characteristic. For the matrix A such a proof cannot apply to all covers of the base graph, and it must fail in characteristic 0 (Remark 7.3); heuristically, it must use the arithmetic of congruence covers.
2. *Characteristic* 2 *and other primes.* Does part (2) fail in every positive characteristic? The case $p = 2$ concerns *Bounded Nesting and Depth Reduction for Matrix Transposition on Tapes*, by the same authors. That paper asks whether binary codes lifted along normal towers of covers with trivial intersection can have positive asymptotic rate; it does not cover 2-adic analytic towers, for example, and notes that a positive limit there would contradict the characteristic-2 case of the conjecture [3, Section 8.1]. Theorem A shows that the conjecture fails in characteristic 3, so the general conjecture no longer supports its characteristic-2 case. For 2-adic analytic towers, whose completions are virtually pro-2, Jaikin-Zapirain's refined conjecture [13] still predicts the universal limit, and Theorem A does not touch it. Goren–Oort configurations at other inert primes, where the supersingular locus is again a union of projective lines indexed by a

Bruhat–Tits tree, are natural candidates in other characteristics. The thirteenth paper of the series proves lower bounds of the kind of Theorem H for the twisted Goren–Oort Tanner codes at every prime $p \geq 7$ inert in $\mathbb{Q}(\sqrt{5})$ and every weight $2 \leq k \leq p-2$ [4, Theorem G]; a failure of part (2) in characteristic p would also need counterparts of the other steps of Theorem A, in particular of Theorem P.

3. *Part (1) along the congruence chain.* Does $\mathrm{rk}_{F/N_k}(A)$ converge? If $\dim_{\mathbb{F}_9} T_{\mathrm{tw}}(X_k) = V_k + c_k - 2$ for every k, which holds when $h^1(\mathcal{O}_{S_k}) = 0$ in characteristic 3 (Section 6.2), then the normalized kernel over $\mathbb{F}_3$ is $22 + 2(c_k - 2)/11^{3k}$, which tends to 22 because the number of cusps c_k grows only like 11^k; the ranks would then converge to 88. The formula holds at $k = 0$, where the kernel is certified (Remark 4.2), and, by computation, at $k = 1$ (Remark 6.11).

4. *The refined conjecture.* Jaikin-Zapirain's refined conjecture for chains whose completion is virtually pro-p [13] is untouched: the bad chain here is pro-11 and the matrix lives in characteristic 3.

5. *Other free groups.* Theorem A concerns the free group of rank 89. Does part (2) fail in characteristic 3 for every non-abelian free group of finite rank, for instance for the free group of rank 2, which contains the free group of rank 89 as a normal subgroup of index 88?

A Index of Lean declarations

The status lines of the paper cite the following declarations, listed by Lean module with their fully qualified names. Module files are relative to the paper's `lean/` directory. The axiom checks in `lean/PaperAxioms.lean` are generated from the same sources.

`LuckApproximation.BaseChange` `Luck.finrank_ker_le_map`, `Luck.quotientMatrix_map`.

`LuckApproximation.Chains` `Luck.Chain.comp`, `Luck.Chain.kerSeq`, `Luck.Chain.rkSeq_eq_sub`, `Luck.not_chainIndependence_of_frequently`, `Luck.not_chainIndependence_of_full`.

`LuckApproximation.ConChecks` `Luck.Con.cosetGraph_ok`, `Luck.Con.labels_ok`, `Luck.Con.spanningTree_ok`, `Luck.Con.splittings_ok`, `Luck.Con.subgroups_ok`.

`LuckApproximation.CyclicTower` `Luck.Op.Ncyc`, `Luck.Op.chiPow`, `Luck.Op.finrank_ker_Ncyc_le`, `Luck.Op.index_Ncyc`, `Luck.Op.isPGroup_Ncyc`, `Luck.Op.rk_Ncyc`.

`LuckApproximation.Hypothesis` `Luck.Op.KernelDensityHypothesis`, `Luck.Op.KernelNotNegligible`, `Luck.Op.kerDensity`, `Luck.Op.kernelNotNegligible_of_density`.

`LuckApproximation.Level1` `Luck.Op.connecting_K`, `Luck.Op.finrank_ker_B0_K`, `Luck.Op.hC`, `Luck.Op.hCinv`, `Luck.Op.hW`, `Luck.Op.hb`, `Luck.Op.hy`.

`LuckApproximation.Main` `Luck.Op.CharThreeChainTarget`, `Luck.Op.ProThreeCofinal`, `Luck.Op.chainIndependence_fails`, `Luck.Op.charThreeChainTarget`, `Luck.Op.charThreeChainTarget_of_density`, `Luck.Op.conjecture_false`, `Luck.Op.exists_chain_tendsto_full`, `Luck.Op.kerDensity_le`, `Luck.Op.kernel_density_tendsto_zero`, `Luck.Op.kernel_density_tendsto_zero_of_luck`, `Luck.Op.luckApproximation_fails`, `Luck.Op.mainTheorem`, `Luck.Op.tendsto_full_of_cofinal`, `Luck.Op.tendsto_full_of_proThreeCofinal`.

LuckApproximation.Monotone Luck.exists_tendsto_of_isPGroup, Luck.rk_le_rk_of_isPGroup.

LuckApproximation.Operator Luck.Op.F, Luck.Op.coupling, Luck.Op.entryEdge, Luck.Op.gamma, Luck.Op.opA.

LuckApproximation.PChain Luck.exists_pChain_le.

LuckApproximation.Sanity Luck.finrank_domain, Luck.rk_eq_sub, Luck.rk_le, Luck.rk_one.

LuckApproximation.Statement Luck.Chain, Luck.Chain.rkSeq, Luck.ChainIndependence, Luck.Convergence, Luck.LuckApproximation, Luck.LuckApproximationConjecture, Luck.phi, Luck.quotientMatrix, Luck.rk.

Also named in the text From this paper's library: Luck.Checks.infinite_of_chain, Luck.Checks.intChain_tendsto, Luck.Checks.rk_one_sub_single, Luck.Checks.rk_zero, Luck.Con.blockNat, Luck.Con.blockNat_eq_cache, Luck.Con.edgeOfNat_eq_cache, Luck.Con.icosians, Luck.Con.label0, Luck.Con.label1, Luck.Con.mono, Luck.Con.red11, Luck.Con.red3, Luck.Literal.equiv, Luck.Literal.nonStrict_imp_luck, Luck.MainTheoremTarget, Luck.Op.opK, Luck.luckApproximation_iff, Luck.not_conjecture_of_target.
From the shared library lean-machine: LeanMachine.CyclicAlg.finrank_ker_le, LeanMachine.CyclicAlg.linIndep, LeanMachine.CyclicAlg.span_eq_top, LeanMachine.KernelCertificate.connecting_injective, LeanMachine.KernelCertificate.finrank_ker_add_le_of_minor, LeanMachine.PackedCheck.prodOK_sound, LeanMachine.ResidualP.exists_goodP_lt, LeanMachine.freeGroup_residually_pGroup.

B Building and verifying the formalization

The Lean project is the directory `lean/` next to this paper. It uses Lean and Mathlib, both at version v4.34.1 (`lean-toolchain`, `lakefile.toml`, `lake-manifest.json`), and requires the repository's shared library `lean-machine` through a relative path, so the sibling directory `lean-machine/` must stay in place. Four modules of `lean-machine` come from this paper: `FreeGroupResiduallyP.lean` (free groups are residually finite p-groups), `PackedMatrixCheck.lean` (kernel checks of matrix identities over $\mathbb{F}_3$ by Kronecker substitution), `KernelCertificate.lean` (kernel bounds from an invertible minor and from an injective connecting map) and `CyclicBandedKernel.lean` (the basis u^k of $K[\mathbb{Z}/N]$ when $u^N = 0$, and banded kernels).

Build. With `elan` installed, run in `lean/`:

```
lake exe cache get && lake build
```

The first command downloads the compiled Mathlib; the second compiles the library, in about 45 seconds on a laptop. The three certificate files `CertB0.lean`, `CertB1.lean` and `CertRest.lean` are checked by `decide +kernel` and build in parallel. No heartbeat limit is changed.

Axioms. The file `PaperAxioms.lean`, generated from the LaTeX sources of this paper by `tools/lean_index.py`, runs `#print axioms` on every declaration named in the paper (Appendix A):

```
lake env lean PaperAxioms.lean
```

Every declaration depends at most on `propext`, `Classical.choice` and `Quot.sound`. No Lean file of the library or of the four modules above uses `sorry`, `admit` or `native_decide`, or declares an `axiom`.

The statement. `Statement.lean` contains the definitions of Section 2. It imports only Mathlib. `Sanity.lean` checks basic properties of the definitions (Lemma 2.2), `Literal.lean` contains the second, independent formalization and its equivalence proof, and `SemanticChecks.lean` the semantic checks. None of these three files is needed by the proof of Theorem A apart from four lemmas of `Sanity.lean`.

The construction and generated data. `Construction.lean` defines the operator from the icosians (Section 3.3); `Operator.lean` assembles `Luck.Op.opA` from it. Three files are generated, and none is trusted: `ConData.lean` (the coset certificate, checked in `ConChecks.lean`), `Data.lean` (packed caches of the incidences and blocks, checked against the construction in `ConCache.lean`; the spanning tree, checked in `ConChecks.lean`; and the voltage χ, a free choice) and `CertData.lean` (the certificate of Lemma 4.1). They are written by

```
python3tools/gen_operator.py
```

run in the paper directory; it needs `numpy` and regenerates the three files byte for byte. It uses `tools/icosian_graph.py`, which builds the icosians, the coset graph and the labels, and `tools/gf.py` for arithmetic in finite fields.

Numerical remarks. `tools/lambda_level1.py` computes the kernel at the first nested level of the congruence chain, with the arguments `untw` for the untwisted datum and `random` s for a random cover with seed s (Remark 6.11); it runs in a few seconds. `python3tools/icosian_graph.py11twuntw` computes the base level, and `python3tools/a5_checks.py` checks the finite group theory of Lemma 6.1 in about half a minute.

Where things are. `Statement.lean` (the conjecture), `Construction.lean`, `ConData.lean` and `Operator.lean` (the matrix), `ConChecks.lean` (Lemma 3.5), `ConCache.lean` and `Bridge.lean` (the caches and the passage to the certificate), `CertChecks.lean`, `CertB0.lean`, `CertB1.lean`, `CertRest.lean` and `Level1.lean` (Lemma 4.1), `CyclicTower.lean` (Lemma 4.3), `Monotone.lean` (Lemma 4.4 and Theorem 4.5), `PChain.lean` (Lemma 4.6), `Chains.lean` (Lemma 5.1 and Proposition 5.2), `BaseChange.lean` (Lemma 5.3), `Hypothesis.lean` (the hypotheses) and `Main.lean` (Theorems P and A and their corollaries).

References

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